



UNIVERSITY OF GREENWICH

Alliance with  Education

COMP1807: COURSEWORK AGILE DEVELOPMENT WITH SCRUM

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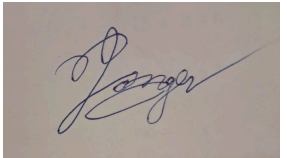
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PART A: Group Report

1. SCRUM Process

1.1 Introducing the Team (Team members and their Roles)

Team member name	Student ID	Individual overall work contribution (%)	Additional notes on task description	Individual Signatures
ĐOÀN TRẦN HUỖNH NHƯ	GCS220086	20%	> SCRUM Master > Distribute workload > Support the team in carrying out the task. > Finalise documentation	
NGUYỄN LÊ THANH ĐẠT	GCS220087	20%	> Development Team > Develop on allocated tasks	
TRỊNH ĐÌNH HIỀN LONG	GCS240217	20%	> Development Team > Develop on allocated tasks	
TRẦN VĂN TẤN	GCS220674	20%	> Development Team > Develop on allocated tasks	
NGUYỄN KHÁNH CHI	GCS240193	20%	> Development Team > Develop on allocated tasks	
Total: 100%				

1.2 Product Vision

The goal is to create a convenient and reliable car hire platform that allows users to easily search, book, and manage their rentals anytime, anywhere

For	Young people, business clients, and existing PrivateHire Cars customers in London.
Who	Need a fast, friendly, and reliable taxi service for city/long-distance trips, airport transfers, or daily chauffeur-driven rentals.
The product name	The PrivateHire online booking system and website.
Product Category	The Private Hire Online Booking System is a GPS-enabled platform optimised for Google Chrome.
That	Enables users to make private bookings anytime, anywhere, with priority over phone bookings, confirmation via text message, and an automatic 15% discount on online transactions. The platform is supported by a highly secure system that ensures all personal and transaction data are protected with advanced encryption.
Unlike	The traditional call centre booking process is currently inefficient, time-consuming, and frequently overloaded with calls.
Our product	provides a fast, friendly, and reliable ride-hailing experience that incentivizes the transition to digital booking with an immediate 15% online discount It integrates advanced safety features, such as the automated text-back service for driver identification, and ensures financial trust through secure Visa Check verification. By tracking popular vehicles and customer loyalty, the system enables personalized special offers that enhance customer satisfaction while optimizing revenue and operational efficiency for PrivateHire Cars

2. Product Backlog

2.1 Turning requirements into user stories.

Req. ID	Requirement List
<i>Account Management & Authentication</i>	
RQ1	Register an account with detailed information, including name, email, address, telephone number, and credit card details.
RQ1	The system will automatically send a confirmation email to the user
RQ5	Customers are first required to log in to their registered account
RQ11	Call Centre staff must be able to follow a verification process to identify callers, allowing them to access existing customer accounts or create new ones for first-time users
<i>Searching & booking Services</i>	
RQ2	The system must enable customers to make enquiries regarding a general trip or a specific journey
RQ3	Book a trip through a phone call
RQ3	Book a trip online with a 15% discount.
RQ4	The system must support various specialized services, including local/long-distance journeys, airport and hotel transfers, "meet & greet" at airports, and driver plus car for daily hire
RQ5	To make an online minicab booking, after logging in, they must provide essential journey details, including the pickup location, destination, date, and time
RQ5	The booking interface must allow users to specify the number of passengers and the required vehicle size, such as a seven-seater minicab
<i>Booking Management & notification</i>	
RQ10	The system must provide an easy way for customers to cancel or amend a booking, as long as the request is made at least 24 hours prior to the journey
RQ6	During the booking process, customers must be able to opt for Email or text-back services; an email is sent once confirmed, and a text message containing the car's registration number, color, and make is sent 10–15 minutes before arrival
<i>Payment</i>	

RQ7	The system must be capable of checking vehicle availability, calculating the total cost, and processing secure payments
RQ7	The platform must accept PayPal and all major credit cards, specifically including American Express
RQ7	Credit card verification must be conducted by the company's bank via a link to the "VISA Check" system.
RQ8	After payment is finalized, the system must generate a printable online receipt that displays the journey details and the final cost
Customer Engagement & Loyalty	
RQ9	Following a journey, customers must be able to leave reviews and star ratings for the trip, the vehicle, and the driver
RQ14	The system must track data regarding the most popular cars and the most loyal customers
RQ14	Based on data built up in customer profiles, the system should automatically send special offer recommendations, which customers can redeem by entering a special offer code during their next booking
Internal Operations & Customer Support	
RQ12	The system must allow Call Centre staff to access real-time information on all available minicabs and drivers to accurately inform customers of prices and availability
RQ13	Call centre staff must be able to interact with customers through the system by replying to queries or sending emails.
RQ13	Sales staff must have the authority to perform administrative actions, such as creating customer accounts, adding new minicabs or drivers, and making bookings

❖ Product Backlog

User story ID	Title	as a	I want to	so that	CR
Account Management & Authentication					
US01	Account Registration and Email Confirmation	New user	I want to register to create an account and receive a confirmation email	so that I can book a car online.	RQ1
US02	Secure	Registered	I want to log in to	so that I can	RQ5

	Account Login	User	my account securely using my credentials.	access booking features and manage my reservations.	
US03	Forgotten Password Recovery	Registered User	I want to retrieve or reset my forgotten password.	so that I can regain access to my account and continue using the booking services.	RQ1
US04	Password Security Update	Registered User	I want to change my password	so that I can update my security credentials.	RQ1
US05	User Profile Management	Registered User	I want to edit my profile information	so that I can keep my personal information up to date	RQ1
US06	Privacy Policy and Data Transparency	Registered User	I want to view privacy policies and data collection details	so that I can understand how my data is collected and used	general manager (PRIVACY)
US07	Customer Identity Verification	Call Centre Staff	I want to follow a verification process	so that we can detect and remove fraudulent profiles to maintain platform trust and safety	RQ11
Searching & booking Services					
US08	Car Service Catalog Browsing	Registered User	I want to browse available car services	so that I can view options before booking.	RQ2
US09	Keyword Based Vehicle Search	Registered User	I want to search for a car by keyword	so that I can quickly find vehicles matching my needs	RQ2
US10	Detailed Service Information and Availability	Registered User	I want to view detailed information about a car service	so that I can understand its features, price, and availability before booking.	RQ2
US11	Category Based	Registered User	I want to filter the car services by	so that I can Narrow down	RQ4

	Service Filtering		category	options quickly.	
US12	Car Price Comparison and Lookup	Registered User	I want to view car prices	so that I can compare options and choose an affordable service.	RQ2
US13	Online Trip Booking with 15% Discount	Registered User	I want to book a trip online	so that I can automatically receive a 15% discount compared to booking via phone	QR3
US14	Phone-Based Trip Booking	Registered User	I want to book my car trip by phoning the call centre	so that I can arrange my transportation even if I lack internet access or prefer assistance from a staff member	RQ3
US15	Daily Hire Service Booking	Registered User	I want to book a driver plus a car for daily hire	so that I can have a dedicated driver and vehicle for my transport needs throughout the day	QR4
US16	Online Trip Booking Details: location, date and time	Registered User	I want to book a minicab online, picking up the location, the destination, the date, and the time of the intended journey	so that ensure a convenient booking experience without the delays of calling the busy Call Centre	QR5
US17	Vehicle Capacity and Size Selection	Registered User	I want to book a car, the number of passengers, and the size of the minicab required, e.g., a seven-seater.	so that I can ensure the assigned vehicle has sufficient capacity for my group, and I can travel to my destination conveniently	QR5
US18	Self-Service FAQ Access	Registered User	I want to access Frequently Asked Questions	so that I can find quick answers to common issues without contacting	RQ2

				support.	
US19	Company Profile and About Us	Registered User	I want to view the "About Us" page	so that I can learn about the company, mission, team, and credibility.	RQ2
Booking Management & notification					
US20	Online Trip Amendment (24-Hour Rule)	Registered User	I want to amend a trip online at least 24 hours prior to the intended journey	so that I can flexibly update my travel details, such as the pickup location and time.	QR10
US21	Booking Cancellation (24-Hour Rule)	Registered User	I want to cancel or amend my booking at least 24 hours prior to the journey	so that I can avoid unwanted trips.	QR10
US22	Text Back Notification for Driver Identification	Registered User	I want to receive a text message 10-15 minutes before arrival with the car's registration number, color, and make	so that I can safely identify my driver and avoid bogus vehicles	QR6
Payment					
US23	Secure Credit Card Payment	Registered User	I want to pay by credit card	so that I can complete my booking.	RQ7
US24	Detailed Price	Registered User	I want to view a detailed price	so that I understand the total cost.	QR7
US25	Payment Verification via VISA Check	Registered User	I want to validate my payment through Verified by Visa check	so that my payment is secure according to the company's bank standards.	QR7
US26	Prices and drivers of existing taxi services	Registered User	I want to access available minicabs and drivers	so that I can provide real-time price and availability information	QR7

US27	Payment Confirmation and Receipt Generation	Registered User	I want to receive a confirmation email and a receipt after payment	so that I have proof of the transaction	QR8
Customer Engagement & Loyalty					
US28	Journey Review and Star Rating	Registered User	I want to provide feedback and a rating	so that I can share my experience and help improve services.	QR9
US29	Membership and Loyalty Upgrades	Registered User	I want to upgrade to a membership plan	so that I can receive discounts on car bookings.	QR14
US30	Promotional Discount Application	Registered User	I want to view and apply promotion/service discounts	so that I can get reduced prices on bookings	QR14
US31	Customer Profiling and Automated Special Offers	admin	I want to track popular cars and loyal customers	so that the system can automatically send special offer recommendations to their profiles	QR14
Internal Operation					
US32	Fleet resource management	Sales staff	I want to add new minicabs or drivers to the system	so that our fleet information remains up to date	QR13
US33	Real-time Fleet Access for Staff	Call Centre Staff	I want to access all available minicabs and drivers	so that they can inform customers accurately about prices and availability	QR12
Support					
US34	Live Chat Incident Support	user	I want to ask for support on an incident via chat	so that I can resolve issues quickly	QR13
US35	Immediate Callback Support (Call Me Now)	Registered User	I want a "Call Me Now" feature	so that I can request an immediate callback from support for urgent issues	QR13

US36	Telephone Incident Support	Registered User	I want to ask for support on an incident via phone	so that I can resolve issues related to my booking quickly.	QR13
US37	Incident and Problem Reporting	Registered User	I want to report a problem	so that I can notify support about issues (e.g., booking errors, vehicle problems, or app bugs).	RQ13
US38	Incident and Problem Management	admin	I want to review and manage reported problems	so that I can resolve user issues related to bookings or services.	RQ13

2.2 Identifying EPICs and breaking them down into smaller user stories.

Methodology and Justification

Epic user stories were identified based on Agile and Scrum principles, which provide an iterative and user-centred approach to requirements development. Within this framework, epics represent high-level features that capture major system functionalities. The team analysed stakeholder requirements and grouped related functionalities into broader categories such as account management, search, booking, and payment. This process followed functional decomposition, structuring requirements from general features into smaller components.

EPIC1: *Account Management & Authentication*

Breakdown into the users' stories.

ID	Story Title (Breakdown)
US01	Account Registration and Email Confirmation
US02	Secure Account Login
US03	Forgotten Password Recovery
US04	Password Security Update
US05	User Profile Management
US06	Privacy Policy and Data Transparency
US07	Customer Identity Verification

Epic 2: Searching & Booking services

Breakdown into the users' stories.

ID	Story Title (Breakdown)
US08	Car Service Catalog Browsing
US09	Keyword-Based Vehicle Search
US10	Detailed Service Information and Availability
US11	Category-Based Service Filtering
US12	Car Price Comparison and Lookup
US13	Online Trip Booking with 15% Discount
US14	Phone-Based Trip Booking
US15	Daily Hire Service Booking
US16	Online Trip Booking Details: location, date, and time
US17	Vehicle Capacity and Size Selection
US18	Self-Service FAQ Access
US19	Company Profile and About Us

Epic 3: Booking *Management & notification*

Breakdown into the users' stories.

ID	Story Title (Breakdown)
US20	Online Trip Amendment (24-Hour Rule)
US21	Booking Cancellation (24-Hour Rule)
US22	text Back Notification for Driver Identification

Epic 4: Payment

Breakdown into the users' stories.

ID	Story Title (Breakdown)
US23	Secure Credit Card Payment
US24	Detailed Price

US25	Payment Verification via VISA Check
US26	Prices and drivers of existing taxi services.
US27	Payment Confirmation and Receipt Generation

Epic 5: Customer Engagement & Loyalty

Breakdown into the users' stories.

ID	Story Title (Breakdown)
US28	Journey Review and Star Rating
US29	Membership and Loyalty Upgrades
US30	Promotional Discount Application
US31	Customer Profiling and Automated Special Offers

Epic 6: Internal Operation

ID	Story Title (Breakdown)
US32	Fleet Resource Management
US33	Real-time Fleet Access for Staff

Epic 7: Support & Communication

Breakdown into the users' stories.

ID	Story Title (Breakdown)
US34	Live Chat Incident Support
US35	Immediate Callback Support (Call Me Now)
US36	Telephone Incident Support
US37	Incident and Problem Reporting
US38	Incident and Problem Management

2.3 User Stories and their Acceptance Criteria

Acceptance Criteria (AC), often referred to as "Conditions of Satisfaction", serve as a concise summary of what a User Story must achieve in order to meet user needs. ACs are negotiable between the Product Owner (PO) and the Development Team to ensure that the implemented features deliver business value while remaining technically feasible.

The purpose of defining AC is to clarify business requirements and specify the conditions that the system must satisfy to meet customer expectations for a given requirement. The presence of AC helps eliminate ambiguity and provides an objective, non-disputable way to determine exactly when a user story can be considered "done".

For a user story to reach completion, all activities—from analysis and development to testing and deployment—must be carried out in such a way that users can successfully execute the scenarios defined in the acceptance criteria.

Below are the acceptance criteria for each user story:

US01: As a new user, I want to register to create an account and receive a confirmation email so that I can book a car online.	
	Acceptance Criteria: 1. A registration form is available with fields for name, email, phone, address 2. All fields have validation (e.g., required, correct format). 3. The password field must contain at least one capital letter, one lowercase letter, and one number. 4. Upon clicking "Submit", the system must send a confirmation email to the registered address and create a new account in the database
US02: As a user, I want to log in to my account securely using my credentials so that I can access booking features and manage my reservations.	
	Acceptance Criteria: 1. The system must allow the user to enter a registered email/username and password to log in. 2. The system must validate the credentials and grant access if the information entered is correct. 3. If the user enters incorrect login details, the system must display an error message and deny access. 4. After successful login, the user must be redirected to their account dashboard, where they can access booking and account management features.
US03: As a user, I want to retrieve or reset my forgotten password so that I can regain access to my account and continue using the booking services.	

	Acceptance Criteria: 1. The "Forgot Password" link must be clearly visible on the login page. 2. The system validates the registered email and sends an automated password reset link 3. The link redirects to a secure page for entering and confirming a new password 4. Display an error for unregistered emails and redirect to the login page after a successful update
US07: As call centre staff, I want to follow a verification process so that we can detect and remove fraudulent profiles to maintain platform trust and safety.	
	Acceptance Criteria: 1. Staff can search for callers by name or telephone number 2. System displays existing profile (address/card details) if found 3. Quick option to create a new account if the caller is new 4. Verification must be completed before making a booking
US09: As a user, I want to search for cars by keyword so that I can quickly find vehicles that match my needs.	
	Acceptance Criteria: 1. Users are to enter keywords in the search bar to search for available cars. 2. The system displays a list of cars that match the entered keyword. 3. If no cars match the keyword, the system displays a message such as "No results found". 4. The search results should be displayed within a few seconds after the user submits the keyword.
US10: As a user, I want to view detailed information about a car service so that I can understand its features, price, and availability before booking.	
	Acceptance Criteria: 1: The system must clearly display the maximum number of passengers each vehicle type can carry. 2: The system must display the available vehicle sizes (e.g., standard or seven-seater). 3: The corresponding price for each vehicle type must be clearly displayed for user comparison. 4: Information about included features (e.g., luggage space and air conditioning) must be fully listed.
US11: As a user, I want to filter car services by category so that I can narrow down my choices quickly.	
	Acceptance Criteria:

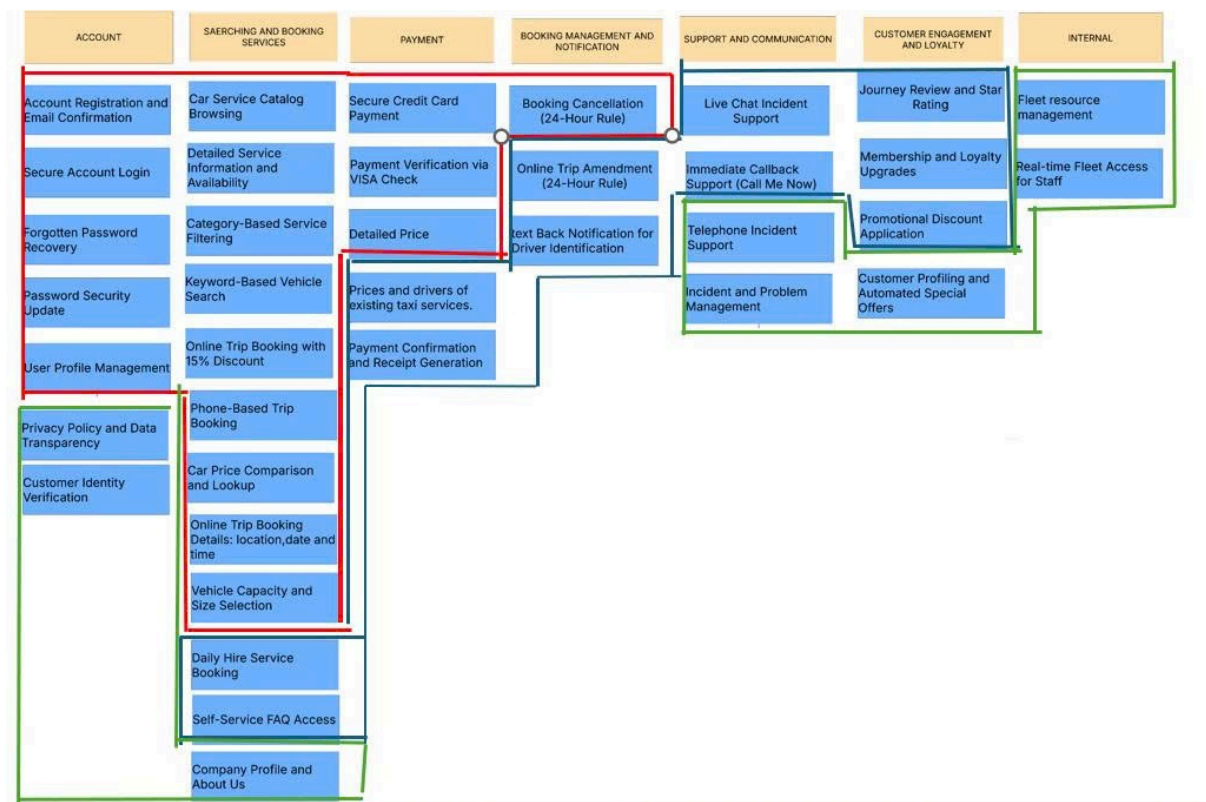
	1. Users can select one or more car service categories (e.g., economy, SUV, luxury, self-drive). 2. The system displays only the cars that match the selected category. 3. Users can clear or change the selected category filters at any time. 4. The filtered results are updated immediately after a category is selected.
US13: As a user, I want to book a trip online so that I can automatically receive a 15% discount compared to booking via phone	
	Acceptance Criteria: 1. 15% discount applies only to online/mobile bookings 2. The payment screen shows "Original Price" vs "Discounted Price". 3. The discount is clearly labelled as "Online Booking Reward". 4. The final discounted total is sent to the receipt and payment gateway.
US21: As a user, I want to cancel my booking at least 24 hours prior to the journey so that I can avoid unwanted trips.	
	Acceptance Criteria: 1. The Cancel or Modify function is only available if performed at least 24 hours before the scheduled departure time. 2. The system must immediately change the booking status to "Cancelled" in the database after cancellation confirmation. 3. Automatic notification: Automatically send confirmation via email or SMS to the customer immediately after cancellation or modification is completed. 4. Allows changes to basic route information and automatically recalculates the actual fare if there are changes.
US23: As a user, I want to pay by credit card so that I can complete my booking.	
	Acceptance Criteria: 1. The payment page must contain a secure form with the following mandatory input fields: Cardholder Name, Credit Card Number, CVV/CVS, and Expiry Date 2. The system must validate that the Credit Card Number follows a standard valid format 3. The CVV field must be restricted to a 3 or 4-digit numeric input 4. If the payment fails (e.g., insufficient funds), a clear error message must be displayed to the user with instructions on how to retry.
US25: As a user, I want to validate my payment through Verified by Visa check so that my transaction is secure and protected from unauthorised use.	
	Acceptance Criteria: 1. The system redirects the user to the Verified by Visa authentication page during payment. 2. The user must enter the required verification code (e.g., OTP) to confirm the transaction. 3. If the verification is successful, the payment is completed successfully. 4. If the verification fails, the system cancels the payment and displays an error message.

US29: As a user, I want to upgrade to a membership plan so that I can receive discounts on car bookings.	
	Acceptance Criteria: 1. The system displays available membership plans and their benefits to the user. 2. The user can select a membership plan to upgrade their account. 3. The system confirms the membership upgrade after the user completes the registration or payment (if required). 4. After upgrading, the system applies membership discounts to eligible car bookings.
US30: As a user, I want to view and apply promotion/service discounts so that I can get reduced prices on bookings.	
	Acceptance Criteria: 1. The system displays available promotions or discount codes to the user. 2. The user can apply a promotion or discount code during the car booking or payment process. 3. The system recalculates the total price after the promotion or discount is applied. 4. The system displays the updated car booking price with the applied discount.
US32: As a sales staff member, I want to add new minicabs or drivers to the system so that our fleet information remains up-to-date.	
	Acceptance Criteria: 1. The system must provide a secure management interface accessible only to authorised Sales staff. 2. The "Add Driver" functionality must include a form with mandatory fields for Full Name, Driver Licence Number, and Telephone Number. 3. The "Add Vehicle" functionality must include mandatory fields for Registration Number, Make, Model, and Colour. 4. The system must validate that the Vehicle Registration Number follows a standard valid format and is unique within the database to prevent duplicates.
US36: As a user, I want to contact support via phone so that I can speak directly with an agent for urgent issues.	
	Acceptance Criteria: 1. The system displays a customer support phone number on the support page (hotline). 2. Users can access the phone contact option easily from the application or website. 3. The system provides clear information about support availability (e.g., support hours). 4. Users can initiate a phone call to the support number from the provided contact details.

2.4 Minimum Viable Product

To effectively manage and prioritise the workload for the private hire car project, the team applied a combination of the Pareto Principle (80/20 rule) and the MoSCoW prioritisation method. The team identified that approximately 20% of the high-level features—namely user account management, car search functionality, booking management, and payment processing—supported by core functionalities such as account registration, secure login, basic car booking with a 15% discount, keyword-based search, booking cancellation, and secure payment would deliver around 80% of the overall business value and operational outcomes.

By focusing on these critical high-level features and their supporting functionalities, the team was able to develop a robust Minimum Viable Product (MVP) that meets the primary needs of stakeholders within the strict six-month deadline while avoiding unnecessary complexity and preventing the development team from being overburdened.



2.5 User stories prioritisation with MOSCOW

MUST	SHOULD	COULD
1.Account Registration and Email Confirmation: (13 points)	1.Phone-Based Trip Booking (1 point)	1.Customer Identity Verification(3 points)
2.Secure Account Login: (5 points)	2.Car Price Comparison and Lookup:(2 points)	2.View policies / data collection(1 point)
3.Car Service Catalog Browsing:(2 points)	3.Payment Confirmation and Receipt Generation : (2 points)	3.Company Profile and About Us(1 point)
4.Category-Based Service Filtering:(2 points)	4. Prices and drivers of existing taxi services:(3 points)	4.Telephone Incident Support (1 point)
5.Category-Based Service Filtering (2 points)	5. text Back Notification for Driver Identification:(5 points)	5.Incident and Problem Management(5 points)
6.Online Trip Booking with 15% Discount(3 points)	5. Live Chat Incident Support (3 points)	6.Customer Profiling and Automated Special Offers(5 points)
7.Keyword-Based Vehicle Search:(2 points)	6.Call me now:(1 point)	7.Fleet resource management (5 points)
8. Detailed Service Information and Availability: (5 points)	7.Journey Review and Star Rating:(3 points)	8.Real-time Fleet Access for Staff (5 points)
9. Secure credit card payment : (5 points)	8. Online Trip Amendment (24-Hour Rule) (2 points)	
10.Booking Cancellation (24-Hour Rule): (2 points)	9.Self-Service FAQ Access(1 point)	
11.Payment Verification via VISA Check: (5 points)	10.Membership and Loyalty Upgrades:(3 points)	
12.Detailed Price: (3 points)	11.Promotional Discount Application(3 points)	
13.Online Trip Booking Details: location,date and time: (2 points)	12.Daily Hire Service Booking(3 points)	
14.Vehicle Capacity and Size Selection: 2points)		
15. Forgotten Password Recovery: (3 points)		
16.Password Security Update: (3 points)		
17. User Profile Management (edit profile): (2 points)		

2.6 User stories estimation using Planning Poker

To address estimation gaps, the team applied Agile Scrum principles, including iterative refinement, team consensus techniques, and continuous feedback, to improve the accuracy

and reliability of story point estimates. Estimation gaps may arise due to uncertainty, varying levels of team experience, or unclear requirements. To reduce these gaps, Planning Poker was used to encourage discussion and achieve consensus among team members. When significant differences in estimates occurred, the team revisited the requirements and further decomposed large user stories into smaller, more manageable tasks to ensure a shared understanding.

❖ **Planning poker**

User story ID	title user story	CR	Story Points	Level of Complexity	Explain story points.
US01	Account Registration and Email Confirmation	RQ1	13	EPIC (high)	Involves designing a complex multi-field form, backend database setup, and SMTP server configuration for automated email confirmations
US02	Secure Account Login	RQ5	5	medium	Requires secure authentication logic, session management, and error handling for unauthorized access
US03	Forgotten Password Recovery	RQ1	3	medium	Moderate effort for implementing secure token-based reset links and automated email triggers
US04	Password Security Update	RQ1	2	medium	Standard internal account management logic with basic input validation
US05	User Profile Management (edit profile)	RQ1	2	medium	Simple CRUD (Update) operation to allow users to modify personal data in the database
US06	Privacy Policy and Data Transparency	general manager (PRIVACY)	1	Low effort	involves displaying static text regarding data transparency and company policies
US07	Customer Identity Verification	RQ11	3	medium	Requires logic for staff to search and verify caller details against existing records.
Searching & booking Services					

US08	Car Service Catalog Browsing	RQ2	2	medium	UI-focused task to display the company's catalog of local/long-distance and transfer services
US09	Keyword Based Vehicle Search	RQ2	2	medium	Implementation of a basic search filter based on user-inputted keywords
US10	Detailed Service Information and Availability	RQ2	5	medium	High UI/Data effort to retrieve and display dynamic attributes: capacity, size, features, and price comparison.
US11	Category Based Service Filtering	RQ4	2	medium	Logical filtering of vehicle results based on service categories.
US12	Car Price Comparison and Lookup	RQ2	2	medium	Backend query to fetch and display standardized pricing for selected routes
US13	Online Trip Booking with 15% Discount	QR3	3	medium	Requires business logic to automatically apply and display the 15% discount for online channels.
US14	Phone-Based Trip Booking	RQ3	1	Low effort	Manual data entry interface for Call Centre staff; minimal system logic required
US15	Daily Hire Service Booking	QR4	3	medium	Specialised service logic for "Driver plus Car" with a different time-based pricing structure.
US16	Online Trip Booking Details: location, date, and time	QR5	2	medium	Standard form validation for pickup/destination locations, dates, and times.
US17	Vehicle Capacity and Size Selection	QR5	2	medium	Logic to ensure vehicle size (e.g., seven-seater) matches passenger requirements
US18	Self-Service FAQ Access	RQ2	1	Low effort	Static information page for common customer inquiries
US19	Company Profile and	RQ2	1	Low effort	Content-based page detailing company history

	About Us				and mission since 2012
Booking Management & notification					
US20	Online Trip Amendment (24-Hour Rule)	QR10	2	medium	Implementation of the 24-hour rule for modifying journey details online
US21	Booking Cancellation (24-Hour Rule)	QR10	2	medium	Automated status update logic restricted by the 24-hour lead time requirement
US22	text Back Notification for Driver Identification	QR6	5	medium	Complex integration with the GPS Despatch System to trigger SMS alerts 10-15 minutes before arrival
payment					
US23	Secure Credit Card Payment	RQ7	5	medium	Security-sensitive task involving secure input forms and validation for card details and CVV
US24	Detailed Price	QR7	3	medium	Dynamic calculation and display of all fare components before final confirmation
US25	Payment Verification via VISA Check	QR7	5	medium	Technical risk involved in integrating a third-party link to the company's bank for VISA Check
US26	Prices and drivers of existing taxi services.	QR7	3	medium	Staff interface to access live price data to inform customers accurately over the phone
US27	Payment Confirmation and Receipt Generation	QR8	2	medium	Generation of printable online receipts summarizing journey and final cost data
Customer Engagement & Loyalty					
US28	Journey Review and Star Rating	QR9	3	medium	Backend logic to capture and link star ratings to specific journeys, vehicles, and drivers

US29	Membership and Loyalty Upgrades	QR14	3	medium	Logic to track customer loyalty and trigger plan upgrades based on usage frequency
US30	Promotional Discount Application	QR14	3	medium	Logic to validate special offer codes and recalculate totals during checkout
US31	Customer Profiling and Automated Special Offers	QR14	5	medium	requires automated data tracking to "push" sales via personalized special offers
Internal Operation					
US32	Fleet resource management	QR13	5	medium	Comprehensive management interface for staff to add/update fleet vehicles and drivers
US33	Real-time Fleet Access for Staff	QR12	5	medium	Requires real-time GPS tracking data visualization for Call Centre staff
Support					
US34	Live Chat Incident Support	QR13	3	medium	Integration of a real-time messaging tool for incident handling
US35	Immediate Callback Support (Call Me Now)	QR13	1	Low effort	Simple "Call Me Now" notification trigger for urgent support
US36	Telephone Incident Support	QR13	1	Low effort	Display of hotline details and support availability hours
US37	Incident and Problem Reporting	RQ13	3	medium	Customer-facing form to report vehicle problems or app bugs
US38	Incident and Problem Management	RQ13	5	medium	Customer-facing form to report vehicle problems or app bugs

Based on Planning Poker estimation using story points (Fibonacci sequence: 1, 3, 5, 8, 13), user stories with high complexity (13 points) were identified as too large to be implemented within a single sprint. These stories were therefore further decomposed into smaller, manageable user stories to ensure they meet Agile requirements for incremental delivery.

EPIC: US01 Account Registration and Email Confirmation (13 point)

ID	Breakdown	Story Points	Explain story points.
US01-1	Registration Form Design & Data Verification	3	This represents a moderate amount of effort focused on the user interface and basic data validation logic. It involves designing the form with fields required by RQ1 (Name, Email, Address, Telephone) and ensuring simple verification like required formats and password security rules.
US01-2	Profile Storage & Database Integration	5	This score reflects higher technical complexity and risk due to the need for a database schema and backend API. It specifically involves the secure handling of sensitive credit card data to meet project security standards and ensuring data integrity across the system.
US01-3	Automatic Confirmation Email System	5	Rated higher due to the uncertainty and integration effort required to configure an SMTP server or external email service. The team must ensure that the "Confirmation Email" trigger is reliable and fires immediately upon successful account creation in the database, requiring rigorous integration testing.

The team applied the Planning Poker technique to estimate the effort required for each user story in the product backlog. Team members voted using story points based on the Fibonacci scale. After discussing differences in estimation, the team reached a consensus on the final story point value.

Below are some of the group's discussion points regarding the poker planning table above.

❖ Planning Poker for Account Registration and Email Confirmation (13 Story Points)

The estimation for the Account Registration and Email Confirmation feature resulted in a rare and immediate 100% agreement across the team. As documented in the SKRUM Planning Poker tool, all members (Tan, Long, Nhu, Dat, and Chi) simultaneously voted for 13 Story Points.



1. Unanimous Voting (The "13 Point" Consensus)

This unanimous decision indicates a perfectly aligned understanding of the feature's high complexity. Unlike other tasks, there was no divergence in the initial round, showing that the team had a shared vision of the effort required for this core security component.

2. Why Did the Team Agree on 13 Points?

The post-vote discussion on Discord highlighted several critical factors that justified this high estimation:

Complex Data Handling: Tan and Dat emphasised that the registration form is the most data-intensive part of the system, requiring multiple layers of validation for personal and credit card details.

Email Service Integration: Long and Chi noted that implementing the SMTP email service for automated confirmations adds significant technical risk, as it involves external service connectivity and asynchronous processing.

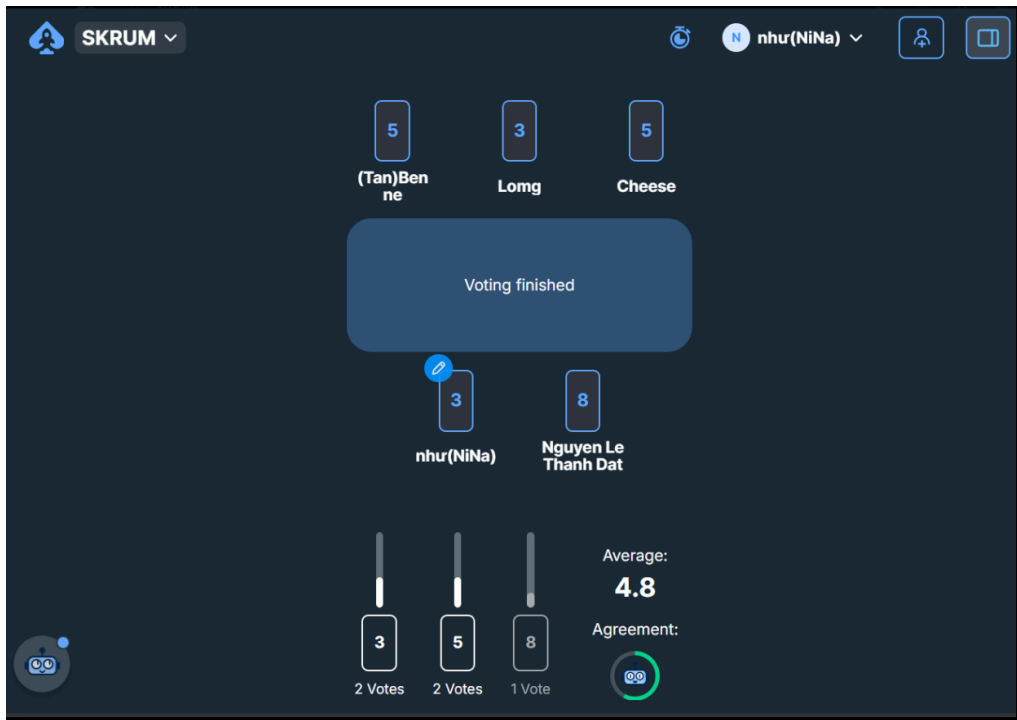
Security Requirements: Nhu pointed out that the "Identity Verification" part of this story (generating and validating OTP codes) is a time-consuming task that must be bug-free to ensure system integrity.

3. Impact on Sprint Planning

Reaching this consensus allowed the team to acknowledge 'Account Registration' as the epic-level task of the sprint. It gave the Scrum Master a clear indication of where the majority of the team's capacity would be invested, ensuring a realistic and achievable sprint epic level.

❖ Planning Poker Case Study: Estimating the Login Feature (5 Story Points)

As the team worked in a distributed environment, we conducted our Planning Poker session through Discord. This allowed for a transparent and documented technical debate regarding the complexity of the "User Login" feature.



SKRUM

nhur(NiNa)

(Tan)Ben ne 5

Long 3

Cheese 5

Voting finished

nhur(NiNa) 3

Nguyen Le Thanh Dat 8

Average: 4.8

Agreement: 100%

2 Votes 2 Votes 1 Vote

Long Yesterday at 9:26 PM
Discussion:

Chi Yesterday at 9:28 PM
I think it is 5 because account login should be highly secured. We should put more effort into it to secure user's security and earn user the most comfortable experience.

Long Yesterday at 9:34 PM
I think the login in would only be 3 because the login user interface and logic processing would not be too complex?

Nhur Yesterday at 9:37 PM
I think 3 points because it simply compares the registered databases to log in; if they match, it will be allowed to log in.

Dat Yesterday at 9:38 PM
I think it's 8 points because when a user logs in, we not only check if the login is correct or incorrect, but also handle password encryption and validate both the email and username. Especially after logging in, maintaining the login session so the user can use other pages is a significant task. The system needs to check the user's role to redirect to the correct dashboard page. If the customer's dashboard is different from the admin's dashboard, the navigation logic will be complicated.

Tan Yesterday at 9:40 PM
I choose 5 points because passwords will be hashed, preventing them from being stored in plain text. The code uses parameterized queries when inserting data, which helps prevent SQL injection. Additionally, when working with the database, sensitive SQL code is only partially shared (about 1/2) to enhance security.

Long Yesterday at 9:41 PM
I see, Dat's point emphasizes the "secure" part of the user story

Chi Yesterday at 9:41 PM
Well, I think that Nhu and Long made a very valid point.

Long Yesterday at 9:44 PM
Tan got the same idea too
So that means Dat thought it would take more effort than Tan for the "secured" part

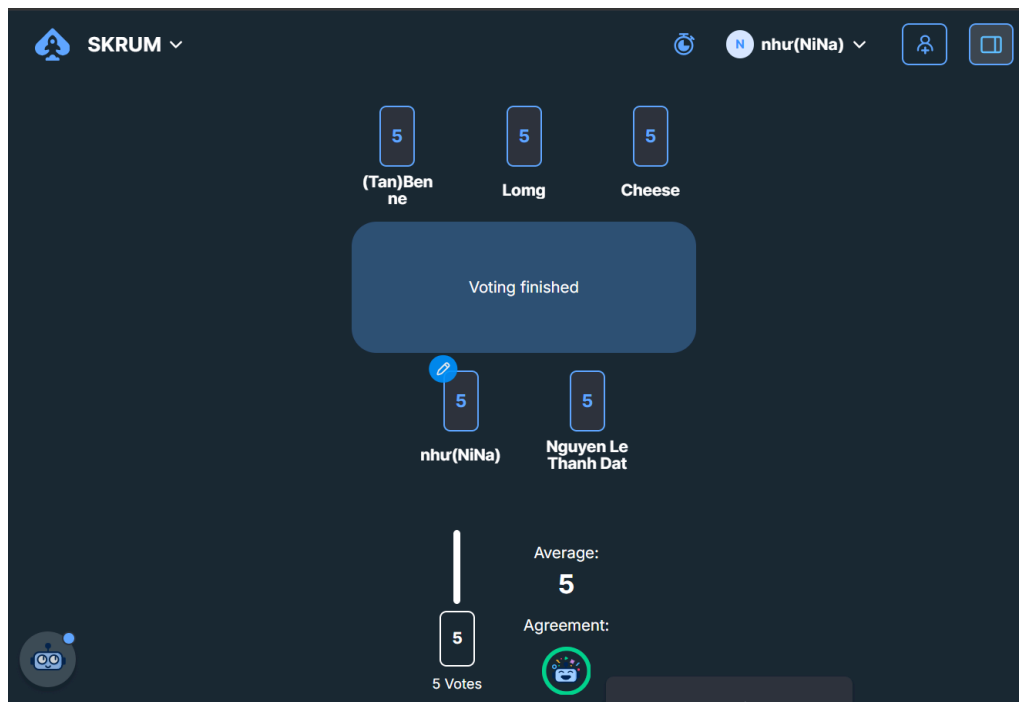
Dat Yesterday at 9:51 PM
Yes, but I think if Tan says it's a 5 out of 10, then I should probably reconsider this part.

Chi Yesterday at 9:55 PM
Well maybe we should vote one more time to see

Dat Yesterday at 9:56 PM
OK

Nhur Yesterday at 9:57 PM
I think 5 points is reasonable.

Long Yesterday at 9:59 PM
I think the Dat and Tan made a really good point, and I lean more on Tan's side. I think there are risks when implementing the secure login function because it's an important function, but even with that in mind, the effort taken shouldn't be as much as 8



1. Initial Divergent Perspectives

The team was initially split between two different technical interpretations of the task:

The "Low Effort" View (Long & Nhu - 3 Points):

Reasoning: They initially focused on the basic functional flow (the "Happy Path"). Their perspective was that the UI is simple and the back-end logic only requires a straightforward comparison between user input and the existing database records.

The "High Complexity" View (DAT – 8 Points):

Reasoning: That raised critical concerns regarding security and navigation. He argued that a "done" login feature must include password encryption, multi-field validation (email/username), and session management. Furthermore, he highlighted the complexity of the redirection logic between the Admin Dashboard and the Customer Dashboard.

2. The Turning Point: Technical Compromise

The discussion shifted when Tan proposed a middle-ground estimate of 5 points. He provided a balanced technical analysis:

Tan's Analysis: He emphasised that while 8 points might be too high for a single feature, a "Secure Login" is not a 3-point task either. It requires parameterised queries to prevent SQL injection and password hashing to protect user data.

Reaching Consensus:

Chi supported Tan's view, noting that 5 points accurately reflects the effort needed for both UI development and essential security implementation.

Dat agreed to lower his estimate from 8 to 5, acknowledging that 5 points sufficiently covers the core security risks he was worried about.

Long and Nhu also revised his and her 3-point vote to 5, admitting that he and she had initially overlooked the non-functional security requirement. "requirements."

3. Final Outcome

After two rounds of voting and technical clarification, the team reached a unanimous agreement to assign 5 story points to the login feature.

Key Takeaway: This exercise proved that what appears to be a simple UI task often involves hidden complexities, such as data security and session integrity, which are vital for a "Potentially Shippable"

3. Sprint One

3.1 Sprint Planning and Sprint Backlog

3.1.1 Sprint Goal

The goal of Sprint 1 is to enable new users to register and log in securely and explore available car services (such as local journeys, airport transfers, and daily hire). Additionally, users can browse the car service catalogue, filter car services by category, and perform keyword-based searches, laying the essential foundation for booking and payment processing in subsequent sprints.

3.1.2 Sprint "Definition of Done".

Definition of Done (DoD) serves as a formal, global checklist that ensures all increments delivered in Sprint 1 are of high quality and ready for a potential release.

While acceptance criteria (AC) are unique to each story, the DoD applies uniformly to every item in the sprint backlog.

For Sprint 1 to be considered successfully completed, the following criteria must be fully satisfied:

- ✓ Acceptance Criteria Met: All specific ACs for the selected user stories (registration, secure login, service browsing, filtering car services by category, and search keyword) have been fully implemented and verified.
- ✓ MVP Readiness: Core features—account registration, secure login, car service catalogue browsing, filtering car services by category, and searching keywords—are functional and deployed in the test environment.
- ✓ Technical Standards: All code has been unit-tested, integrated into the shared repository, and refactored to ensure high quality and zero technical debt.
- ✓ Scrum Board Accuracy: All user stories in the sprint backlog have been moved to the "DONE" column after verification.
- ✓ Stakeholder Verification: The product owner (Mr Andrew Fowler) has reviewed and approved the potentially shippable increment for the demo.

3.1.3 Sprint Backlog

❖ Sprint Capacity Planning

Step 1: Key Variables	
	Team Size: 4 developer
	Sprint Length: 2 weeks (10 working days, excluding weekends)

	allocation: 3 Full-time developer (8h/day) and 1 Part-time developer
	Individual Availability: 100% available during scheduled hours
Step 2: Daily Capacity Calculation	
	Full-time developer 1: 8 hours/day
	Full-time developer 2: 8 hours/day
	Full-time developer 3: 8 hours/day
	Part-time developer 4: 4 hours/day
	Total Daily Team Capacity: $(3 \times 8h) + (1 \times 4h) = 28 \text{ hours/day}$
Step 3: Total Sprint Capacity	
	Formula: (Daily Team Capacity) $\times$ (Number of Working Days) Calculation: 28 hours/day $\times$ 10 days = 280 man-hours.
	including 40 hours allocated for Scrum activities such as sprint planning, daily Scrum, sprint review, and retrospective. Productive Time: 280-40 = 240 hours.
	Velocity per hour = Total Daily Team Capacity / 10 working days = 28 / 280 man-hours = 28 / 280 = 0.1
	Forecasted capacity: 240 $\times$ 0.1 = 24 story points

Derived from the productive time analysis, the team's forecasted capacity for this sprint is 24 story points. To match this capacity, the team carefully selected the below stories for the sprint backlog, totalling about 24 points. This consistency between the projected capacity and the actual workload ensures a realistic, non-overloaded sprint plan, thereby maximising the likelihood of delivering all committed work meeting the Definition of Done (DoD) by the sprint's conclusion.

User story ID	Title (user story)	Story Points	CR
US01-1	Registration Form Design & Data Verification	3	RQ1

US01-2	Profile Storage & Database Integration	5	RQ1
US01-3	Automatic Confirmation Email System	5	RQ1
US02	Secure Account Login	5	RQ1
US08	Car Service Catalog Browsing	2	RQ2
US09	Keyword-Based Vehicle Search	2	RQ2
US11	Category-Based Service Filtering	2	RQ4
Total: 24 points			

3.1.4 Breaking down user stories included in the Sprint Backlog into Tasks

Having finalised the sprint backlog with a committed capacity of 24 story points, the team proceeded to break down these user stories into actionable technical tasks. The following section details the task breakdown required to fulfil the acceptance criteria and definition of done for each selected item.

Title (User story)	Task Description	Task Category	Estimates (hours)
US01: Account Registration and Email Confirmation	US01-1: Registration Form Design & Data Verification		
	US01-1.1 Design wireframes for registration	Design	6
	US01-1.2 Develop logic for data validation (email and phone number)	code	4
	US01-1.3 implement secure password rules (uppercase, lowercase, numbers)	code	4
	US01-2 : Profile Storage & Database Integration		
	US01-2.1 Database Integration for storing user profiles	code	8

	US01-3: Automatic Confirmation Email System		
	US01-3.1 SMTP server config & email security triggers	Research	4
	US01-3.2 Implement backend logic for automated confirmation email triggers	code	7
	US01: Functional and security testing for the full registration process. (For the whole US01-1, US01-2, US)	test	6
US02: Secure Account Login	US02.1 Secure password hashing (BCRYPT)	Research	4
	US02.2 Design the login page interface.	Design	4
	US02.3 Implement credential validation and session management	code	8
	US02.4 Handle error messages and dashboard redirection	code	4
	US02.5 Security testing for login authentication	test	5
US08: Car Service Catalog Browsing	US08.1 SQL query optimization for car data	Research	4
	US08.2 Design UI to display service categories (Airport, long-distance...)	Design	7
	US08.3 Query service data from the database to display	code	5
	US08.4 Test display compatibility on mobile and web.	test	4
US09: Keyword-Based Vehicle Search	US09.1 SQL query optimization for car data	Research	5
	US09.2 Design the search bar and results display area	Design	4
	US09.3 Develop a keyword-based search algorithm	code	6
	US09.4 Handle the display status when there are "No results found"	code	6

	US09.5 Test search query speed performance	test	4
US11: Category-Based Service Filtering	US11.1 AJAX/Fetch for real-time filtering	Research	5
	US11.2 UI/UX Design for Service Catalog	Design	6
	US11.3 Develop Service Data API	code	7
	US11.4 Frontend Service Grid Implementation	code	6
	US11.5 Functional Testing for Browsing	test	5
Total: 138 hours			

❖ **Justification for Task Estimation and Agile Thinking**

For Sprint 1, the team committed to 24 story points based on our projected capacity. Following Scrum.org principles, we avoid 100% resource utilisation to focus on maximising value from complex problem-solving. Out of 240 total hours, 111 hours are allocated to core technical tasks mentioned above. The remaining 102 hours serve as "Slack time" for rigorous XP engineering disciplines:

- Pair Programming (35 hrs): Collaborative development of complex security logic for US01 and US02.
- Refactoring Mercilessly (25 hrs): Continuously optimising code to eliminate technical debt.
- Strategic Buffer (12 hrs): Reserved for resolving impediments while maintaining an energized, sustainable pace

Test Tracking List (30 hours)

Test Case ID	Test Type	Acceptance Criteria (AC)	Expected Result	Actual Result	Status	Notes	Test Date
TC01	UI/UX	US01-1.1 Design wireframes for registration	The user interface layout is compatible with the specifications of many website models.	As Expected	Pass		17/02

TC02	Unit	US01-1.2 Develop logic for data validation (email and phone number)	System triggers error messages for invalid email or phone number formats.	As Expected	Pass		18/02
TC03	Security	US01-1.3 implement secure password rules (uppercase, lowercase, numbers)	The system rejects weak passwords and requires uppercase, lowercase, and numbers.	Not As Expected	Fail	Additional notes are needed, such as requiring passwords to include uppercase and lowercase letters and special characters, so users know how to set the correct password. Fixed.	18/02
TC04	Database	US01-2.1 Database Integration for storing user profiles	Registered user data is accurately stored in the MySQL database.	As Expected	Pass		19/02
TC05	Functional	US01-3.2 Implement backend logic for automated confirmation email triggers	A confirmation email is automatically sent upon successful registration.	As Expected	Pass		20/02
TC06	UI/UX	US02.2 Design the login page interface.	The login fields and buttons are precisely positioned for	As Expected	Pass		23/02

			a user experience on website.				
TC07	Unit	US02.3 Implement credential validation and session management	Users can only log in with verified accounts and valid credentials.	As Expected	Pass		23/02
TC08	Functional	US02.4 Handle error messages and dashboard redirection	Displays "Invalid credentials" on failure; redirects to Dashboard on success.	As Expected	Pass		24/02
TC09	UI/UX	US08.2 Design UI to display service categories (Airport, long-distance, Hotel,...)	Categories for Airport, Hotel, and Long-Distance services are clearly displayed.	Not As Expected	Fail	Some services were missing, including airport services and hotel services. Fixed.	25/02
TC10	Integration	US08.3 Query service data from the database to display	Display the exact basic services retrieved from the database.	As Expected	Pass		26/02
TC11	UI/UX	US09.2 Design the search bar and results display area	The search bar is accessible, and the results grid responds well across website.	As Expected	Pass		26/02
TC12	Logic	US09.3 Develop a keyword-based	Results accurately match the input keyword	As Expected	Pass		27/03

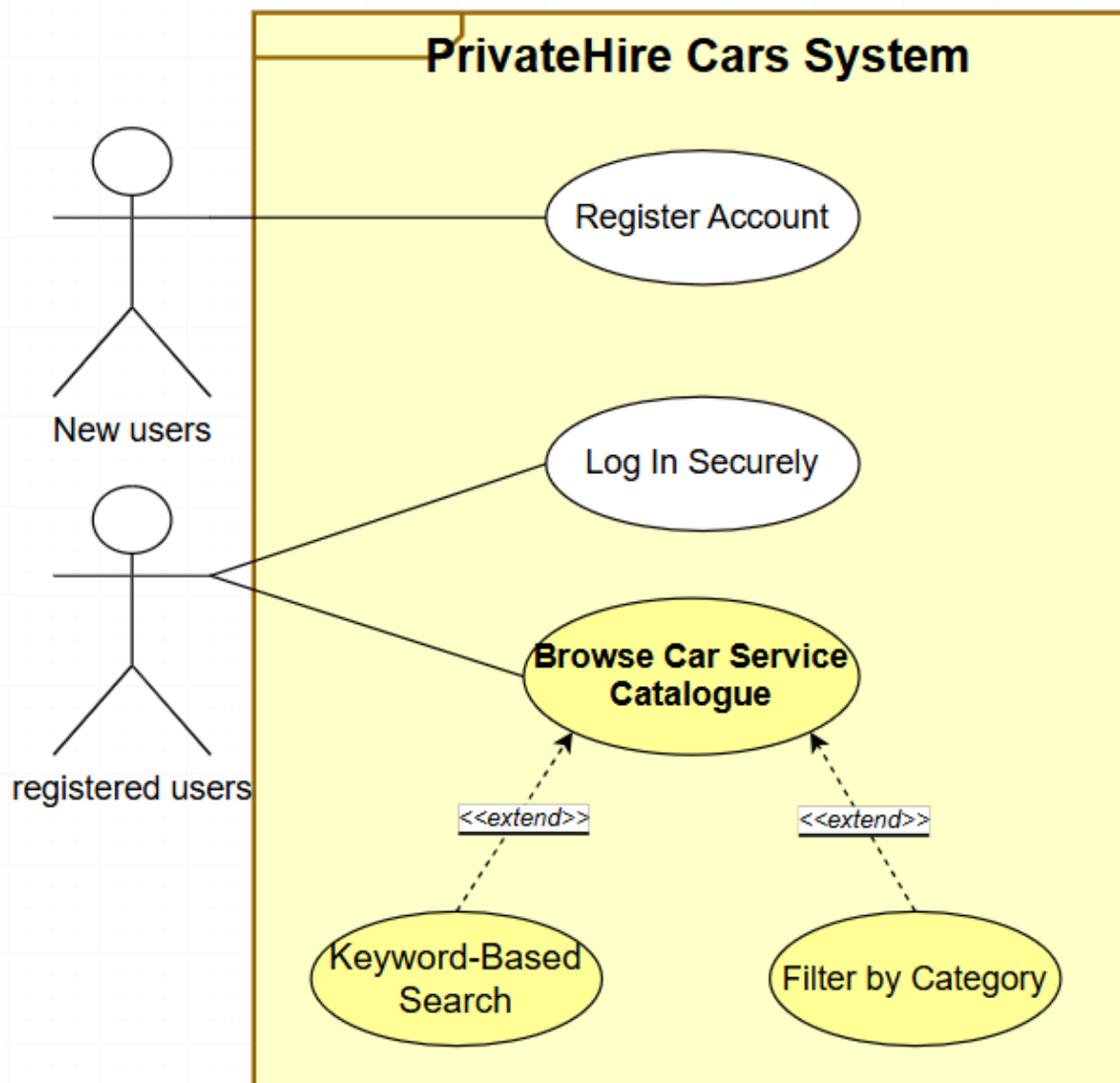
		search algorithm	(e.g., "Toyota", "Honda").				
TC13	Usability	US09.4 Handle the display status when there are "No results found"	Clearly notifies the user when no car matches the search query.	As Expected	Pass		27/03
TC14	UI/UX	US11.2 UI/UX Design for Service Catalog	The catalogue layout provides a "Fast, Friendly, and Reliable" feel.	As Expected	Pass		02/03
TC15	API	US11.3 Develop Service Data API	The API returns the correct MySQL data structure to the admin page for vehicle services.	Not As Expected	Fail	No APIs are being used; instead, MySQL is being used to transfer data. Fixed.	02/03
TC16	Database	US11.4 Frontend Service Grid Implementation	The service grid displays price breakdowns and discounts correctly.	As Expected	Pass		03/03

3.2 Sprint Cycle

3.2.1. Modelling

As part of Sprint 1, our team created a use case diagram to visualise the key functionalities we planned to deliver. These included allowing users to register an account, log in securely, browse the car service catalogue, filter car services by category, and search for vehicles by keyword. The diagram helped us better understand how users would interact with the system and provided a clear reference for our development work. The modelling process ensured that we were all aligned on what needed to be built and that our design accurately reflected the requirements set out in the selected user stories.

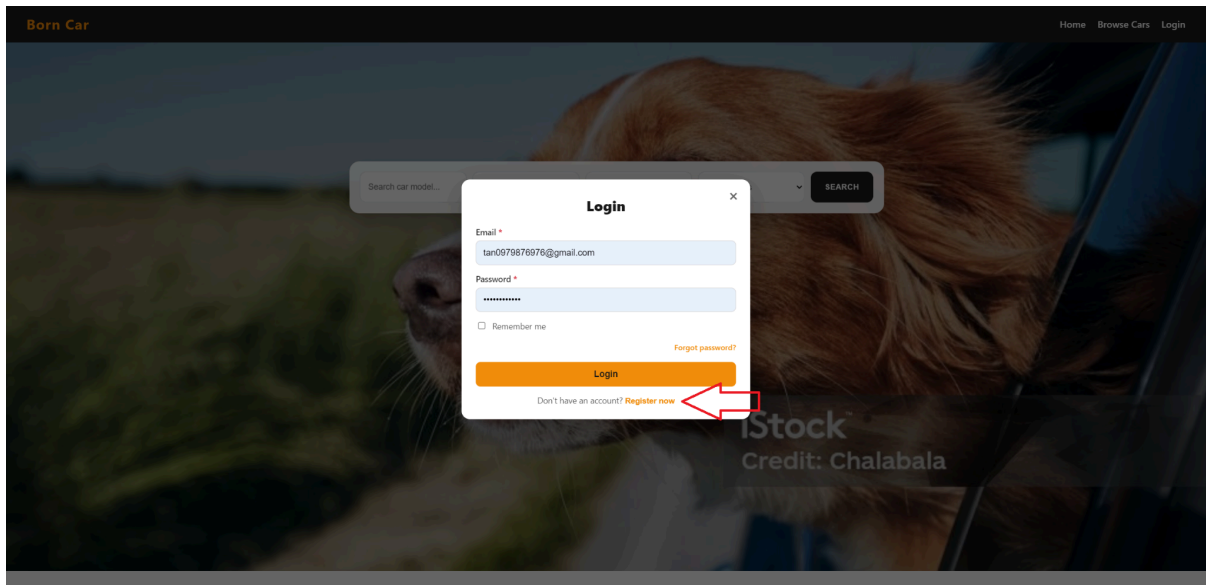
3.2.1.1. Use-Case Diagram



3.2.2 Development

3.2.2.3 US01 – Account Registration and Email Confirmation

User Story: As a new user, I want to register for an account and receive a confirmation email so that I can access the platform and book a car online



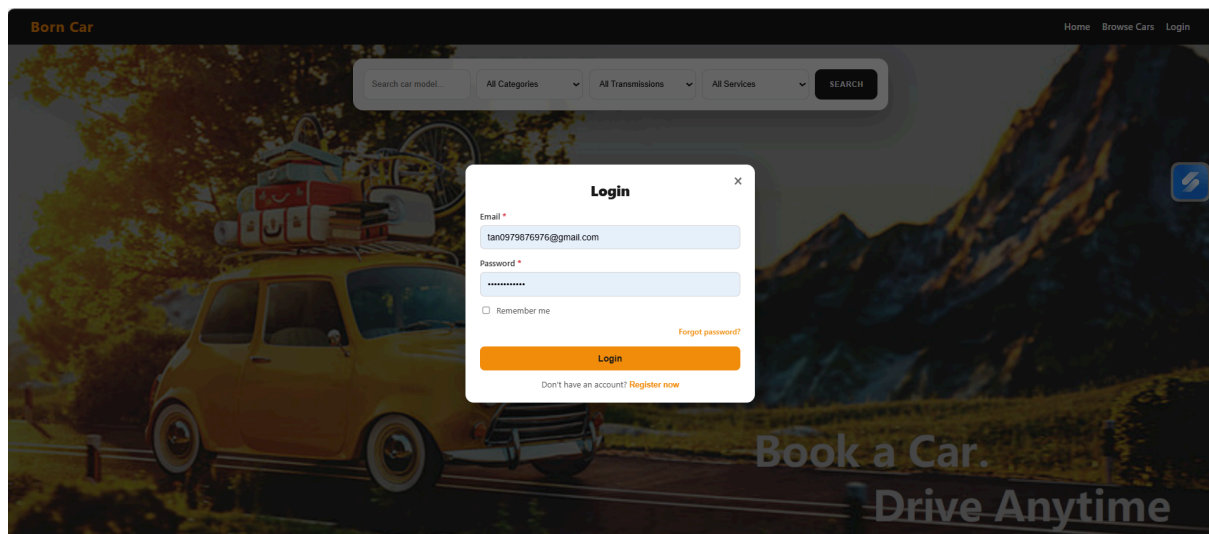
After clicking the login button, a window like this will appear. If the user does not have an account, click the register button next to the arrow.

Register	Register
Full name * Tran Van Tan	Full name * Thanh Đạt
Email * tan0979876976@gmail.com	Email * tan0979876976 ! Please include an '@' in the email address. 'tan0979876976' is missing an '@'
Phone number * 0979876976	03636739091111
Address 47/5H Phường Xuân Thới Đông, HCM, VN	Address Vũng Tàu
Password * *****	Password * *****
Register	Register
Cancel	Cancel
Already have an account? Login	Already have an account? Login

After clicking register, the registration form will appear. Users must fill in the characters in the fields marked with a red asterisk. A red asterisk indicates a required field. The full name can be anything, the email address must include @gmail.com, the phone number must be of sufficient length, and the password must contain at least one special character, one uppercase letter, and both letters and numbers.

3.2.2.4 US02 – Secure Account Login

User Story: As a user, I want to log in to my account securely using my credentials so that I can access booking features and manage my reservations



After registering an account, the login interface will appear.

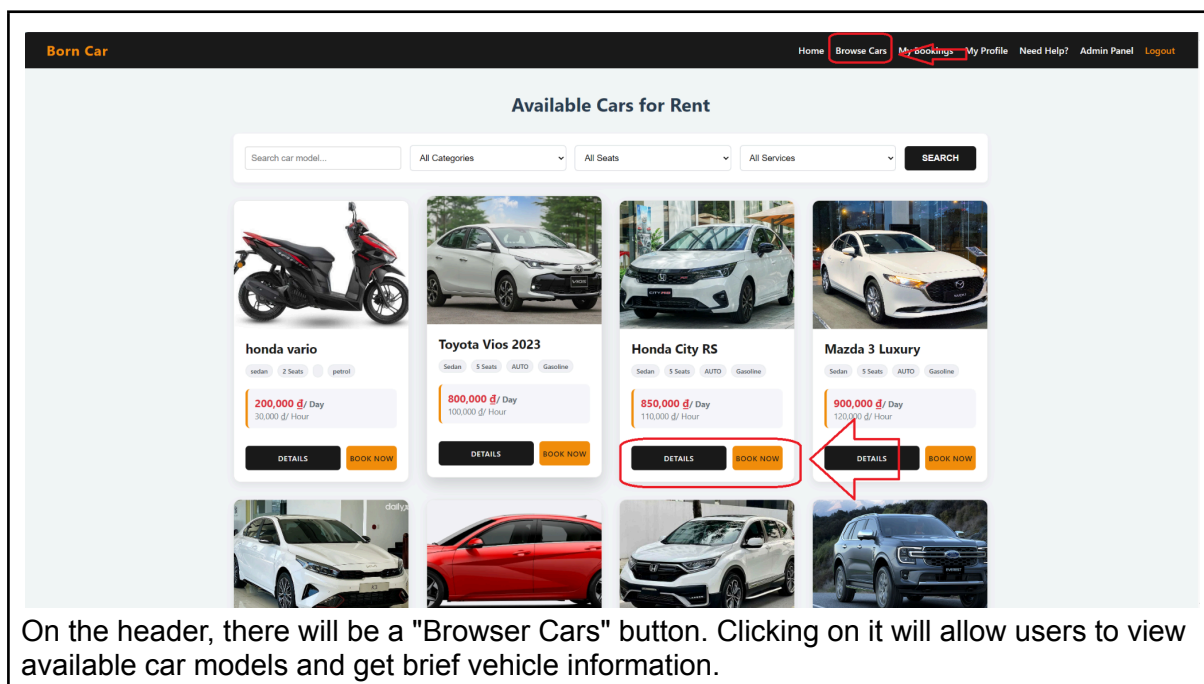
Or if you already have an account, you can click the login button in the header.

Users must enter the correct password and email address used during registration.

If incorrect, login will be denied and a message will appear. The email or password may be incorrect.

3.2.2.5 US08 – Car Service Catalog Browsing

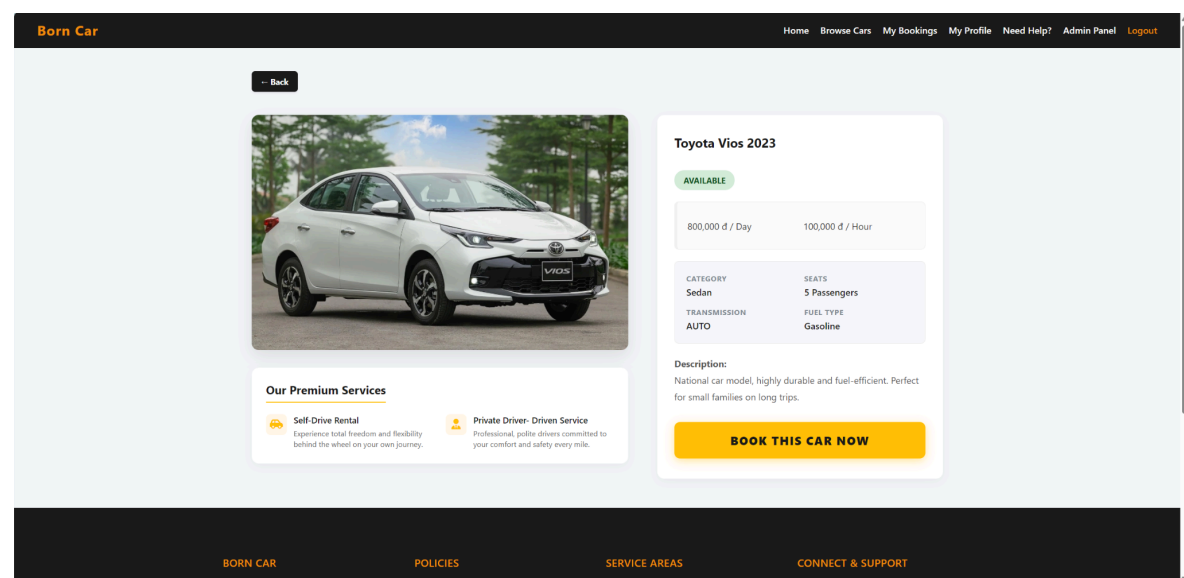
User Story: As a user, I want to browse available car services so that I can view options before booking



On the header, there will be a "Browser Cars" button. Clicking on it will allow users to view available car models and get brief vehicle information.

If users want to see more details, there is a "Details" button for more information about the car and services.

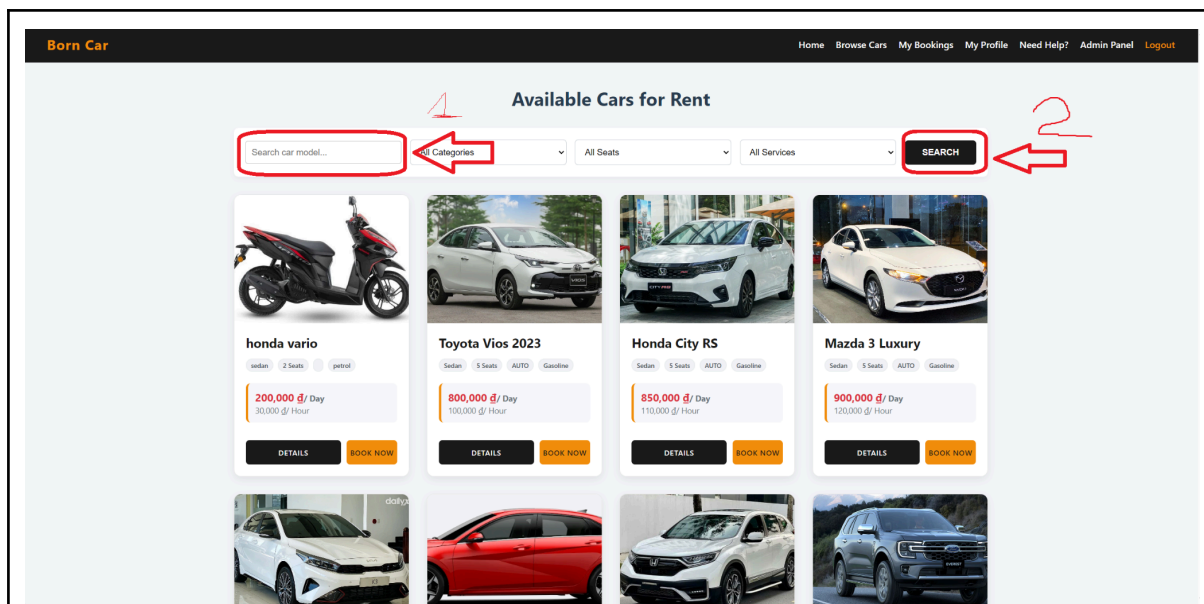
The "Book Now" button allows users to skip directly to the booking step, bypassing the step where they need to specify the vehicle they want to book.



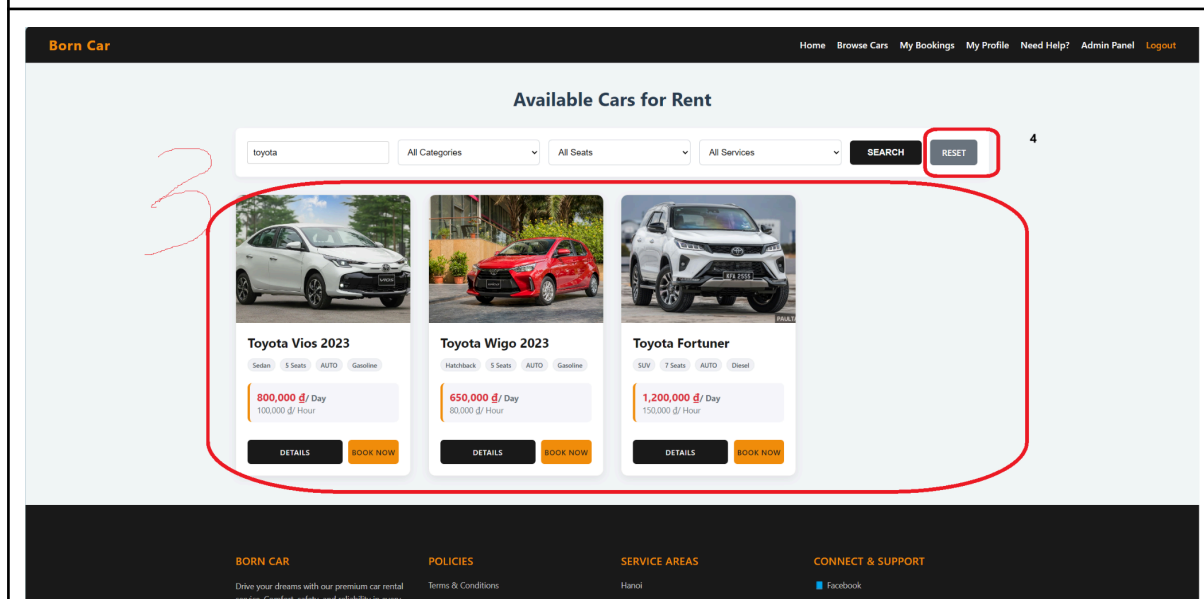
This is the interface after clicking the "detail" button, helping users to better understand the vehicle's information.

3.2.2.4 US09 – Keyword-Based Vehicle Search

User Story: As a user, I want to search for cars by keyword so that I can quickly find vehicles that match my needs



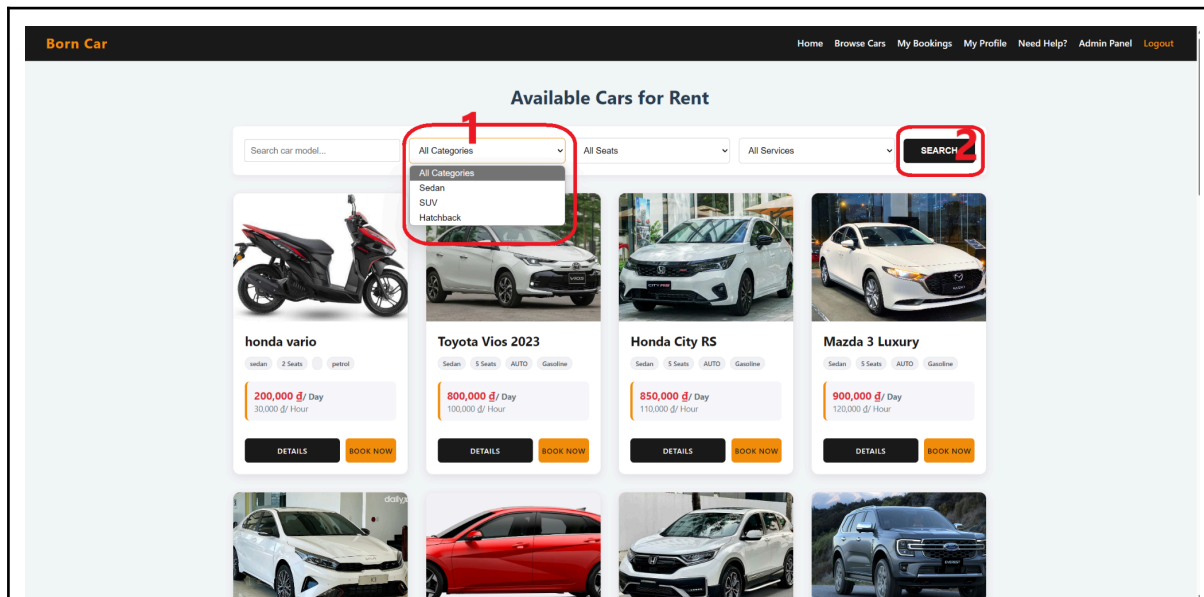
First, enter the keywords into the search box.
Press the search button.
A list of vehicles matching the keyword below will appear.



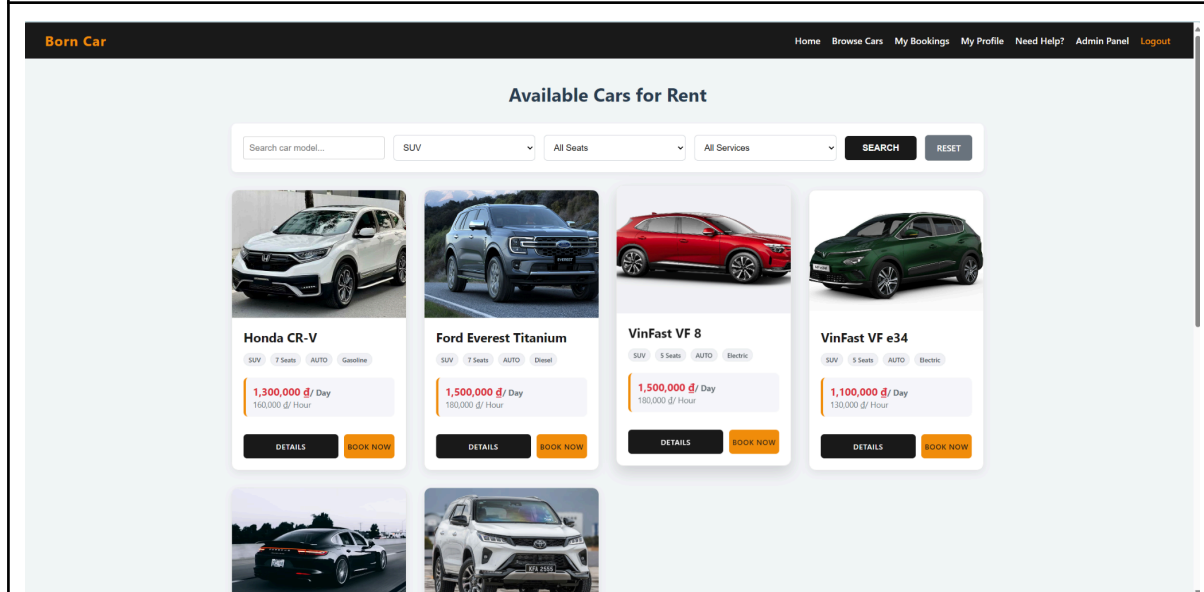
This is what the screen looks like after entering a keyword and clicking the search button.
After searching, you can press the reset button to reset the search bar.

3.2.2.6 US11 – Category-Based Service Filtering

User Story: As a user, I want to filter car services by category so that I can narrow down my choices quickly



To filter categories, users must click the "All Categories" button to display the list, then select a category and click the "Search" button.



This is the list of vehicles after filtering the categories to SUVs.

3.2.2.7 Scrum Board

Monitoring progress is an important aspect that can greatly influence any project's successfulness. In SCRUM Agile, this monitoring is done through a SCRUM Board - a visual display that tracks projects in either a physical or virtual format. For this reason, many tools have been created and put into practice to help with managing this SCRUM Board, such as Trello, Visual Paradigm. Our team decided to use Jira to manage our progress, since it is the tool we are most familiar with, and because it has sufficient and advanced features to help with our monitoring.

The SCRUM Board is updated just before the Burndown Chart and right after each Daily SCRUM meetings by team members so that there is minimal forgetfulness of the meeting content.

For this section, the report will display the SCRUM Board while the Sprint is progressing and when the Sprint is concluded to show that it has been used to measure advancements

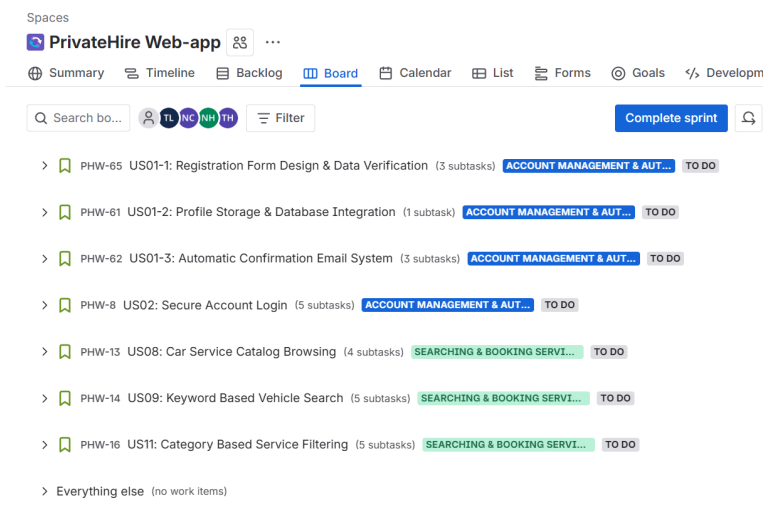


Figure 3.2.2.7.1: Sprint 1 SCRUM Board before the development process when showing only user stories

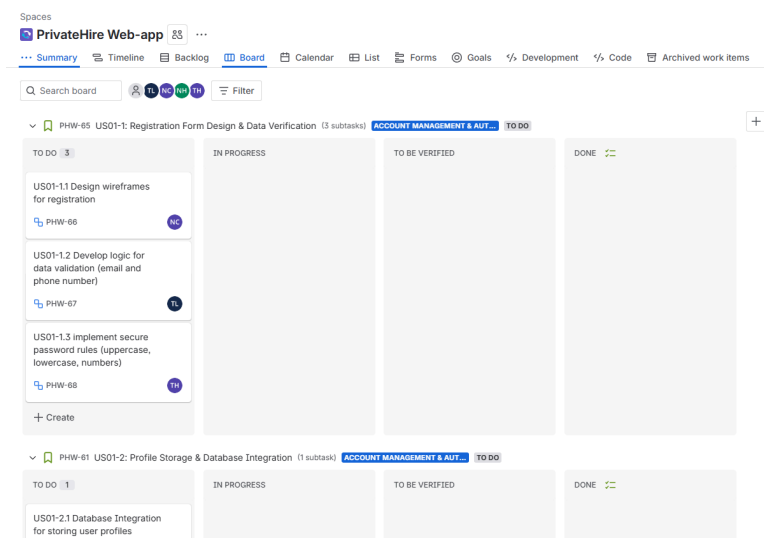


Figure 3.2.2.7.2: Sprint 1 SCRUM Board before the development process when showing tasks

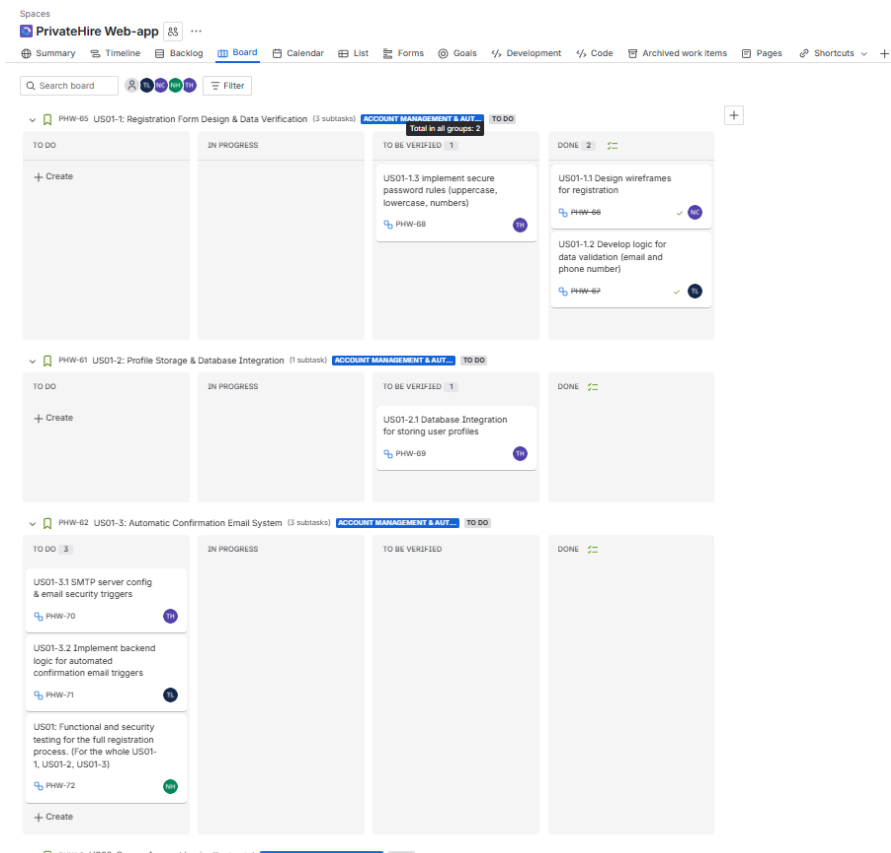


Figure 3.2.2.7.3: Sprint 1 SCRUM Board during the development process

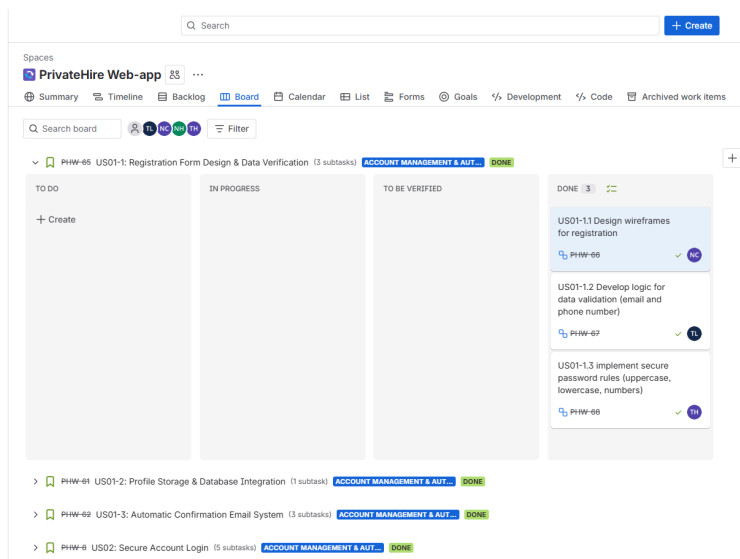


Figure 3.2.2.7.4: Sprint 1 SCRUM Board after the Sprint has concluded

3.2.2.8 Burndown charts

The burndown chart is an effective way for the team to follow the progress of the Sprint. Following the straight line of progression, the team tries to complete as much work as possible before or after the weekends, as the team's line naturally rises during such days. This burndown chart is updated after every Daily SCRUM and after updates on the SCRUM Board for accurate monitoring

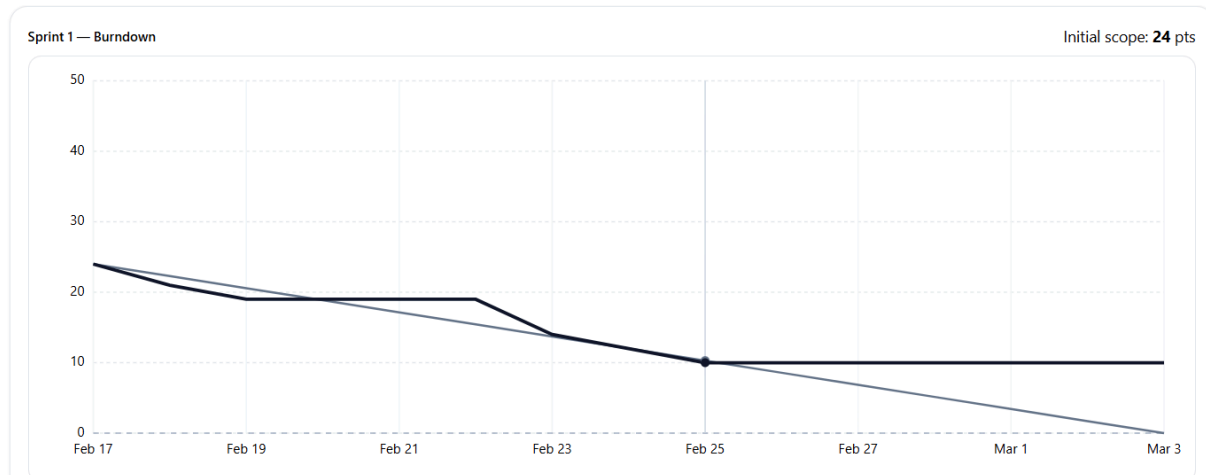


Figure 3.2.2.8.1: Sprint 1's Burndown chart up to February 25th

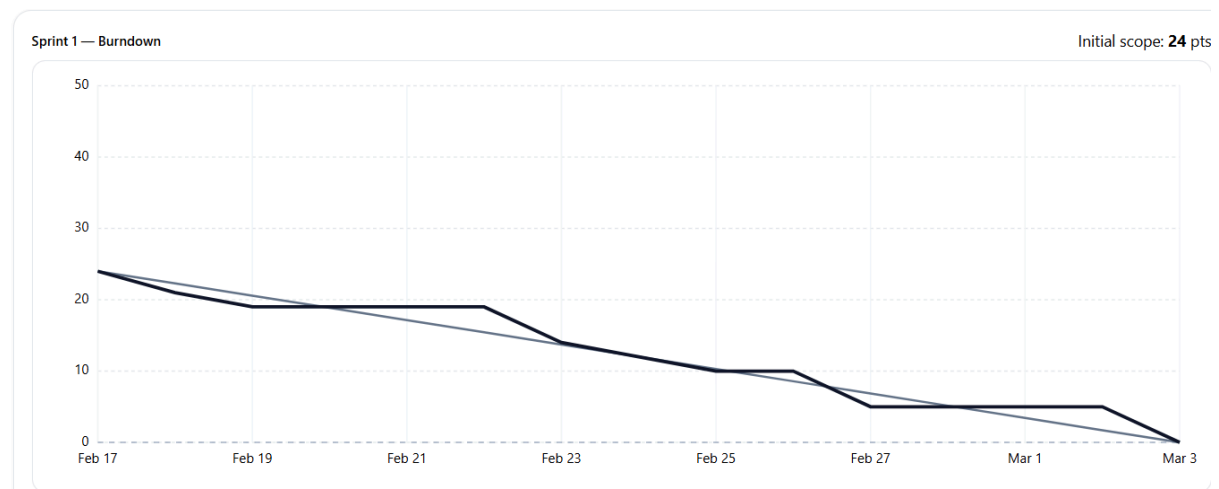


Figure 3.2.2.8.2: Sprint 1's Burndown chart at the end of the Sprint on March 3rd

3.2.2.9 Meetings

❖ Daily meetings

Day	Focus Areas	Artifact status
-----	-------------	-----------------

1-2	UI/UX Design for US01, US02, and US08.	Scrum Board: Design tasks moved to <i>Done</i> .
4-5	Core Backend & DB Integration (US01.4).	Burndown: Dropped from 210h to 160h.
7-8	Search Algorithm (US09.2) & API Development.	Scrum Board: US08 and US09 reach 100% completion.
9-10	Final Security & Functional Testing.	All backlog items moved to the 'Done' column.

❖ **Sprint review meetings**

To improve system usability and prevent user errors, the following enhancements were implemented:

1. Registration Form (US01)

Password Guidance: Added a UI hint: "Password must include uppercase, lowercase, special characters, and numbers."

Mandatory Fields: Implemented red asterisks (*) to clearly distinguish required fields from optional ones.

Benefit: Reduces registration friction by providing clear input requirements upfront.

2. Search Functionality (US09)

Input Validation: Restricted the use of special characters (e.g., %, @) in the search bar.

Benefit: Prevents potential system errors and ensures search queries remain relevant to car services.

❖ **Retrospective**

At the end of Sprint 1, the team conducted a retrospective meeting to reflect on our workflow, identify what went well, and pinpoint areas for improvement to ensure a smoother execution for Sprint 2 (on booking and payment).

1. **What Went Well**

- Technical Synergy: The integration between the WinForms UI and the SQL database was successful and stable.
- Communication: Daily Scrum meetings helped the team identify blockers early, especially regarding the SMTP configuration for US01.
- UX Focus: Proactively adding "usability hints" (password rules and red asterisks) improved the overall quality of the increment.

2. **What Didn't Go Well**

- Estimation Accuracy: Some complex tasks in US01 (registration logic) took longer than initially estimated, causing a slight delay in the middle of the sprint.
- Code Standardisation: Initial code merges had some naming convention conflicts, which required extra time for refactoring.

3. **Improvements for Sprint 2**

Buffer Management: Refine the story point estimation by breaking down complex user stories into smaller, more manageable sub-tasks.

Documentation Standards: Standardise naming conventions for UI controls and database tables to minimise refactoring time.

4. Team Performance Summary

Overall, the team performed effectively in Sprint 1, delivering a potentially shippable increment that includes a working UI and backend foundation. The Scrum Master maintained high visibility of the progress through the Scrum board. All developers contributed to implementation and peer reviews, feeling better prepared for Sprint 2, which will introduce complex transaction and payment logic

4. Sprint Two

4.1 Sprint Planning and Sprint Backlog

4.1.1 Sprint Goal

The objective of Sprint 2 is to enable users to complete the full booking process and secure payments. Building upon the registration and login foundation established in Sprint 1, this sprint implements the complete "business life cycle" flow. This includes entering essential journey details (pickup location, destination, date, and time), online trip booking with a 15% discount; selecting vehicle capacity and size, automatically calculating total costs (detailed service information and availability); integrating secure credit card payments with mandatory bank verification via the VISA Check system, and online trip cancellation/amendment (24-hour rule).

4.1.2 Sprint "Definition of Done"

In the Scrum framework, the Definition of Done (DoD) is a shared understanding within the team of the quality standards required for a product increment to be considered complete.

- ✓ It ensures that every feature delivered at the end of the sprint meets both the technical and business expectations of PrivateHire Cars.
- ✓ For Sprint 2, the following criteria must be fully satisfied for the Sprint to be considered successfully completed:
 - ✓ Core Business Cycle Completion: The full online booking and payment flow is implemented, including the logic for calculating the total cost (detailed service information and availability) and automatically applying the 15% online booking discount and the 24-hour cancellation/amendment rule.
 - ✓ User Story Fulfilment: All seven User Stories in the Sprint Backlog (US10, US13, US16, US17, US21, US23, US25, and US27) are finished and have passed their specific acceptance criteria.
 - ✓ MVP Enhancement: The Minimum Viable Product (MVP) is updated to handle the complete journey from entering location/date/time details to processing secure payments and automatically applying the 15% online booking discount and the 24-hour cancellation/amendment rule.
 - ✓ Comprehensive Testing: All features are verified in the test environment, including journey data validation, the 24-hour cancellation/amendment rule, and simulated VISA Check verification and the automated 15% discount logic for online bookings.

- ✓ Regression Testing: The team has confirmed that core features from Sprint 1 (such as registration and secure login) remain fully functional after integrating the new Sprint 2 codebase
- ✓ Refactoring and Documentation: The code has been mercilessly refactored to ensure a simple, maintainable design, and all associated technical documentation or manuals have been updated
- ✓ Quality and UX Review: The team has verified the end-to-end user experience, ensuring a seamless flow from car selection to the final transaction
- ✓ Product Owner Approval: The product owner (or tutor representing stakeholders) has reviewed and accepted the deliverables, declaring them ready for the prototype demonstration.

4.1.3 Sprint Backlog

❖ Sprint Capacity Planning

Step 1: Key Variables	
	Team Size: 4 developer
	Sprint Length: 2 weeks (10 working days, excluding weekends)
	allocation: 3 Full-time developer (8h/day) and 1 Part-time developer
	Individual Availability: 100% available during scheduled hours
Step 2: Daily Capacity Calculation	
	Full-time developer 1: 8 hours/day
	Full-time developer 2: 8 hours/day
	Full-time developer 3: 8 hours/day
	Part-time developer 4: 4 hours/day
	Total Daily Team Capacity: $(3 \times 8h) + (1 \times 4h) = 28 \text{ hours/day}$
Step 3: Total Sprint Capacity	
	Formula: (Daily Team Capacity) × (Number of Working Days) Calculation: 28 hours/day × 10 days = 280 man-hours.
	including 40 hours allocated for Scrum activities such as sprint planning, daily Scrum, sprint review, and retrospective. Productive Time: 280-40 = 240 hours.

	Velocity per hour = Total Daily Team Capacity / 10 working days = 28 / 280 man-hours = 28 / 280 = 0.1
	Forecasted capacity: 240 x 0.1 = 24 story points

Derived from the productive time analysis, the team's forecasted capacity for this sprint is 24 story points. To match this capacity, the team carefully selected the below stories for the sprint backlog, totalling about 24 points. This consistency between the projected capacity and the actual workload ensures a realistic, non-overloaded sprint plan, thereby maximising the likelihood of delivering all committed work meeting the Definition of Done (DoD) by the sprint's conclusion.

User story ID	Title (user story)	Story Points	CR
US10	Detailed Service Information and Availability	5	RQ2
US13	Online Trip Booking with 15% Discount	3	RQ3
US16	Online Trip Booking Details: location, date, and time	2	RQ5
US17	Vehicle Capacity and Size Selection	2	RQ5
US21	Online trip cancellation/amendment (24-Hour Rule)	2	RQ10
US23	Secure Credit Card Payment	5	RQ7
US25	Payment Verification via VISA Check	5	RQ7
Total: 24 points			

4.1.4 Breaking down user stories included in the Sprint Backlog into Tasks

Having finalised the sprint backlog with a committed capacity of 24 story points, the team proceeded to break down these user stories into actionable technical tasks. The following section details the task breakdown required to fulfil the acceptance criteria and definition of done for each selected item.

Title (User story)	task Description	task category	estimates (hours)
--------------------	------------------	---------------	-------------------

US10: Detailed Service Information and Availability	US10.1 GPS real-time availability integration	GPS real-time availability integration	2
	US10.2 Design UI for vehicle details and feature list	Design	5
	US10.3 Develop backend logic to fetch real-time price and features	code	4
	US10.4: Verify data accuracy for vehicle availability	test	2
US13: Online Trip Booking with 15% Discount	US13.1 Online booking channel identification logic	Research	5
	US13.2 Design a "Payment Summary" interface that displays the original price and the "Online Booking Reward" label.	Design	4
	US13.3 Program the logic to calculate the 15% discount for orders triggered from online/mobile channels.	code	7
	US13.4 Integrate the discounted price data into the payment gateway system and the receipt generation module	code	6
	US13.5: Test the functionality to confirm the 15% discount is accurate and the label displays as required.	test	3
US16: Online Trip Booking Details: location, date, and time	US16.1 Address validation API integration	Research	4
	US16.2 Create booking form with mandatory journey fields	Design	3
	US16.3 Implement validation for pickup/destination	code	5

	address		
	US16.4 Integration test for booking data submission	test	5
US17: Vehicle Capacity and Size Selection	US17.1 Capacity constraint logic for various sizes	Research	4
	US17.2 Update UI to include vehicle size dropdown	Design	3
	US17.3 Implement backend filtering by passenger capacity	code	5
	US17.4 Test capacity constraints logic (min/max passengers)	test	2
US21: Online Trip Cancellation/Amendment (24-Hour Rule)	US21.1 Secure card encryption & CVV rules	Research	2
	US21.2 Design UI for booking cancellation buttons	Design	5
	US21.3 Develop logic to verify the 24-hour cancellation rule	code	5
	US21.4 Implement status update (to "Cancelled") in the database	code	4
	US21.5 Code fare recalculation logic for trip amendments	code	4
	US21.6 Test automatic email/SMS notification for cancellation	test	5
US23: Secure Credit Card Payment	US23.1 Secure card encryption & CVV rules	Research	5
	US23.2 Design secure credit card input form	Design	4
	US23.3 Implement card format and CVV numeric validation	code	5

	US23.4 Integrate payment gateway for transaction processing	code	5
	US23.5 Test failure scenarios (e.g., insufficient funds)	test	4
US25: Payment Verification via VISA Check	US25.1 Research VISA Check API and Bank integration requirements	Research	4
	US25.2 Implement secure redirection logic to the bank's authentication page	code	6
	US25.3 Handle callback response and OTP verification status from the bank	code	6
	US25.4 Create unit and integration tests for success and failure (e.g., wrong OTP) scenarios	test	5
Total: 138 hours			

❖ Justification for Task Estimation and Agile Thinking

The team committed to 24 story points based on past velocity, focusing on value delivery rather than 100% resource utilisation. Out of 240 total hours, we allocated 111 hours for core technical tasks and 102 hours as "Slack time" to implement essential XP and Scrum disciplines.

- Pair Programming (35 hrs): Collaborative coding on complex modules like US25 (VISA Check) to improve quality and share knowledge.
- Refactoring Mercilessly (25 hrs): Continuous optimisation of code to eliminate technical debt.
- Strategic Buffer (12 hrs): Reserved for resolving unexpected impediments while maintaining an energised, sustainable pace.

Test Tracking List (30 hours)

Test Case ID	Test Type	Acceptance Criteria (AC)	Expected Result	Actual Result	Status	Notes	Test Date
TC17	UI/UX	US10.2	Vehicle	As	Pass		06/03

		Design UI for vehicle details and feature list	features and details are displayed clearly on website.	Expected			
TC18	Unit	US10.3 Develop backend logic to fetch real-time price and features	Accurate pricing and feature data are retrieved from MySQL.	As Expected	Pass		06/03
TC19	UI/UX	US13.2 Design a "Payment Summary " interface that displays the original price and the "Online Booking Reward" label.	The 15% discount label and original price are visible to the user.	Not As Expected	Fail	Discount labels are not showing. Fixed.	09/03
TC20	Unit	US13.3 Program the logic to calculate the 15% discount for orders triggered from online/mobile channels.	The total price is correctly reduced by exactly 15% for online bookings.	Not As Expected	Fail	The discount amount was incorrect compared to the request. Fixed.	09/03
TC21	Functional	US13.4 Integrate the discounted price data into	The final receipt and payment gateway reflect the discounted	Not As Expected	Fail	The receipt only shows the original	09/03

		the payment gateway system and the receipt generation module	price.			price, missing the discounted price. Fixed.	
TC22	UI/UX	US16.2 Create booking form with mandatory journey fields	Form includes required fields for location, date, and time.	As Expected	Pass		10/03
TC23	Logic	US16.3 Implement validation for pickup/destination address	System prevents identical pickup and destination entries.	Not As Expected	Fail	This logic doesn't work. Fixed.	10/03
TC24	UI/UX	US17.2 Update UI to include vehicle size dropdown	User can select vehicle size (e.g., 7-seater) from a dropdown menu.	As Expected	Pass		11/03
TC25	Functional	US17.3 Implement backend filtering by passenger capacity	Only vehicles meeting the passenger count are shown in results.	As Expected	Pass		11/03
TC26	UI/UX	US21.2 Design UI for booking cancellation buttons	Cancellation buttons are accessible and clear in the booking history.	As Expected	Pass		12/03
TC27	Unit	US21.3 Develop	Cancellation is blocked	Not As Expected	Fail	Users can	13/03

		logic to verify the 24-hour cancellation rule	if the journey is less than 24 hours away.			cancel at any time. Fixed.	
TC28	Database	US21.4 Implement status update (to "Cancelled") in the database	Database reflects the correct "Cancelled" status for the trip.	As Expected	Pass		16/03
TC29	Unit	US21.5 Code fare recalculation logic for trip amendments	Total fare is recalculated correctly when trip details are amended.	As Expected	Pass		16/03
TC30	UI/UX	US23.2 Design secure credit card input form	Input form for credit cards is clean and mimics standard payment UI.	As Expected	Pass		17/03
TC31	Unit	US23.3 Implement card format and CVV numeric validation	System rejects non-numeric CVVs or invalid card formats.	As Expected	Pass		18/03
TC32	Integration	US23.4 Integrate payment gateway for transaction processing	System communicates successfully with the payment processor.	As Expected	Pass		19/03
TC33	Functional	US25.2 Implement secure redirection	Users are redirected to the external	Not As Expected	Fail	Users are not redirected to any	19/03

		n logic to the bank's authentication page	bank page for OTP entry.			external banking pages to enter the OTP for authentication. Fixed.	
TC34	Vulnerability	US25.3 Handle callback response and OTP verification status from the bank	System processes bank approval/denial and updates payment status.	As Expected	Pass		20/03

4.2 Sprint Cycle

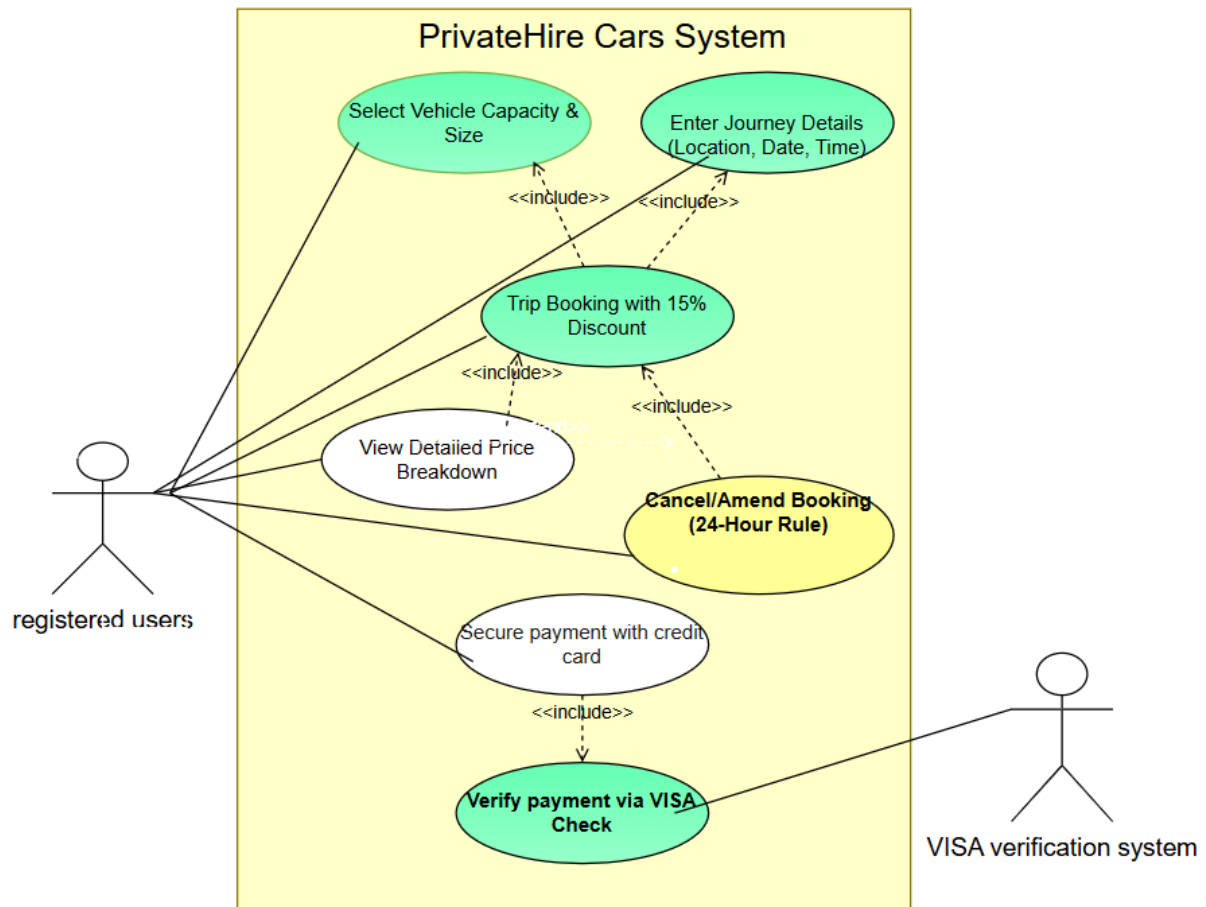
4.2.1. Modelling

The team has expanded the use case diagram to reflect the transition from basic browsing to full booking and service purchasing capabilities. This modelling phase centres on a seamless checkout experience for private hire car customers. Key functionalities include entering online trip booking details (location, date, and time), in-app booking with integrated 15% discount logic; viewing detailed price breakdowns; and executing secure payments via the Visa Checkout system.

The updated diagram ensures team alignment on the purchase workflow and visualises critical dependencies. For instance, it illustrates the requirement to specify passenger counts and vehicle types (e.g., 7-seaters) before proceeding to checkout, as well as the application of discounts prior to order confirmation. This supports sprint planning by making the end-to-end transactional logic transparent for both developers and testers, ensuring all business rules—including the 24-hour cancellation policy—are fully accounted for.

Adhering to Agile principles, the diagram provides 'just enough' structure to guide implementation without the burden of redundant documentation. This model was rigorously reviewed during sprint planning and served as a constant reference throughout development, ensuring every implemented feature aligns perfectly with a defined user story and its corresponding use case.

4.2.1.1. Use-Case Diagram



4.2.2 Development

4.2.2.1 US10 – Detailed Service Information and Availability

User Story: As a user, I want to view detailed information about a car service so that I can understand its features, price, and availability before booking

Booking Preview

Kia K3 Premium
 Rate (Day): 850,000 d
 Rate (Hour): 110,000 d

PICK-UP TIME
08 Apr 2026 - 15:00

RENTAL DURATION
2 Days

PICK-UP AREA
Binh Thanh District

DROP-OFF TIME
10 Apr 2026 - 09:00

SERVICE TYPE
Chauffeur (With Driver)

PICK-UP LANDMARK
landmark

Back
dailyxe

Kia K3 Premium

AVAILABLE

850,000 d / Day 110,000 d / Hour

CATEGORY	SEATS
Sedan	5 Passengers
TRANSMISSION	FUEL TYPE
AUTO	Gasoline

Description:
Youthful, dynamic, and fully loaded with options. Excellent soundproofing and deep AC cooling.

BOOK THIS CAR NOW

Our Premium Services

Self Drive Rental
Experienced total freedom and flexibility behind the wheel on your own journey.

Private Driver- Driven Service
Professional, polite drivers committed to your comfort and safety every mile.

RENTAL DURATION 2 Days PICK-UP AREA Binh Thanh District Car Rental Fee 1,700,000 đ Driver Fee (If applicable) 1,000,000 đ Subtotal 2,700,000 đ TOTAL PAYMENT 2,700,000 đ Got a Voucher? -- Select a Voucher -- ← Back CONFIRM & GO TO PAYMENT	
---	--

Similar to other types in the search bar, just click on the character and then press the search button.

To delete the search bar, there is a reset button.

4.2.2.2 US13 – Online Trip Booking with 15% Discount

User Story: As a user, I want to book a trip online so that I can automatically receive a 15% discount compared to booking via phone

PICK-UP AREA Binh Thanh District	PICK-UP LANDMARK landmark
-------------------------------------	------------------------------

Car Rental Fee	1,700,000 đ
Driver Fee (If applicable)	1,000,000 đ
Subtotal	2,700,000 đ
Discount (WELCOME15)	- 405,000 đ

TOTAL PAYMENT 2,295,000 đ

Got a Voucher?

WELCOME15 - Discount 15.00%
-- Select a Voucher --
WELCOME15 - Discount 15.00%
LOYAL20 - Discount 20.00%
VIPMAX25 - Discount 25.00%

← Back

User can click got a voucher for choose voucher

4.2.2.3 US16 – Online Trip Booking Details (location, date, and time)

User Story: As a user, I want to provide essential journey details including the pickup location, destination, date, and time, so that I can ensure a convenient booking

Booking Preview



Kia K3 Premium
Rate (Day): 850,000 đ
Rate (Hour): 110,000 đ

PICK-UP TIME
08 Apr 2026 - 15:00

DROP-OFF TIME
10 Apr 2026 - 09:00

RENTAL DURATION
2 Days

SERVICE TYPE
Chauffeured (With Driver)

PICK-UP AREA
Binh Thanh District

PICK-UP LANDMARK
landmark

RENTAL DURATION
2 Days

SERVICE TYPE
Chauffeured (With Driver)

PICK-UP AREA
Binh Thanh District

PICK-UP LANDMARK
landmark

Car Rental Fee	1,700,000 đ
Driver Fee (If applicable)	1,000,000 đ
Subtotal	2,700,000 đ

TOTAL PAYMENT **2,700,000 đ**

Got a Voucher? -- Select a Voucher --

[Back](#) [CONFIRM & GO TO PAYMENT](#)

4.2.2.4 US17 – Vehicle Capacity and Size Selection

User Story: As a user, I want to indicate the number of passengers and the required vehicle size so that I can ensure the assigned vehicle has sufficient capacity

Available Cars for Rent

All Categories

7 Seats

All Services

SEARCH

RESET



Honda CR-V
SUV 7 Seats AUTO Gasoline
1,300,000 đ/ Day
160,000 đ/ Hour
[DETAILS](#) [BOOK NOW](#)



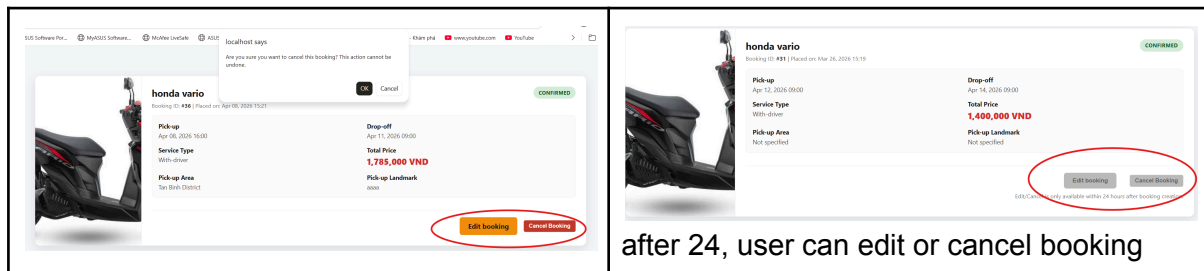
Ford Everest Titanium
SUV 7 Seats AUTO Diesel
1,500,000 đ/ Day
180,000 đ/ Hour
[DETAILS](#) [BOOK NOW](#)



Toyota Fortuner
SUV 7 Seats AUTO Diesel
1,200,000 đ/ Day
150,000 đ/ Hour
[DETAILS](#) [BOOK NOW](#)

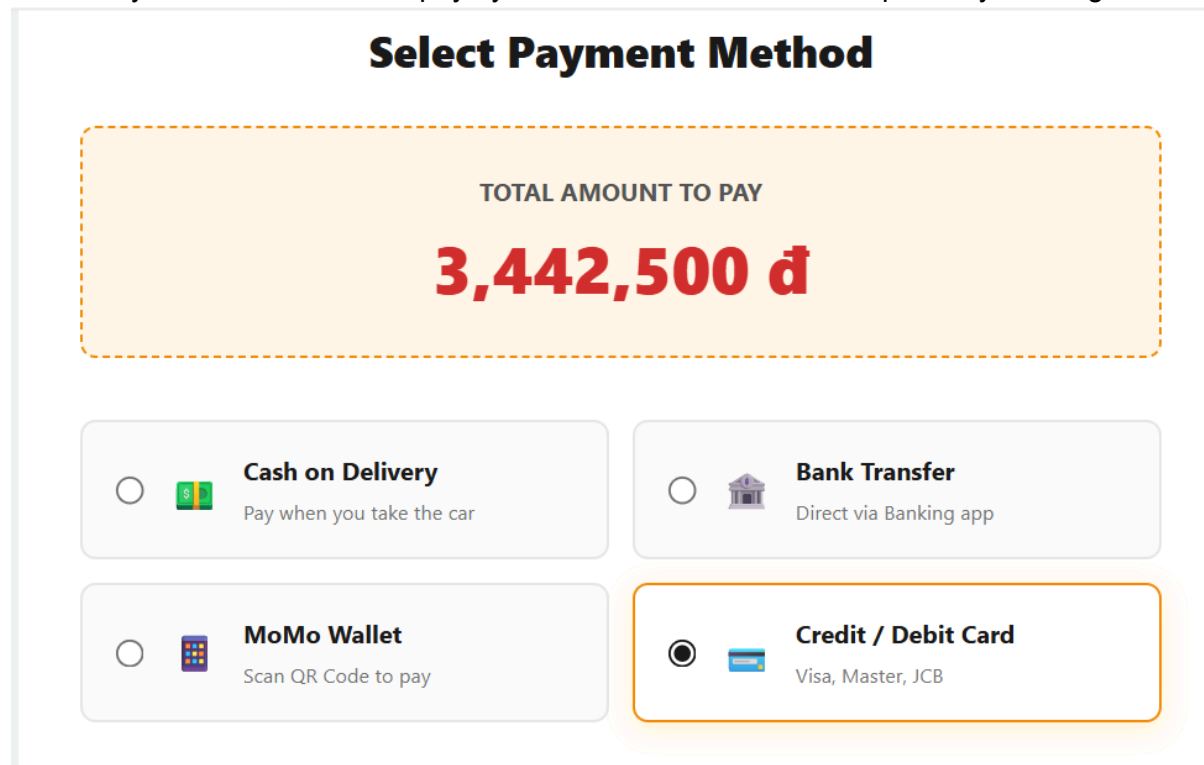
4.2.2.5 US21 – Online Trip Cancellation/Amendment (24-Hour Rule)

User Story: As a user, I want to cancel or amend my booking at least 24 hours prior to the journey so that I can avoid unwanted trips



4.2.2.6 US23 – Secure Credit Card Payment

User Story: As a user, I want to pay by credit card so that I can complete my booking



4.2.2.7 US25 – Payment Verification via VISA Check

User Story: As a user, I want to validate my payment through VISA Check so that my transaction is secure according to bank standards

Secure Card Details

Cardholder Name

tran van tan

Credit Card Number

4441 4411 1111 1111

Expiry Date (MM/YY)

12/30

CVV

123

For demo mode: card numbers ending with 0000 simulate insufficient funds.

CONFIRM & PAY NOW

✓ 100% Secure & Encrypted Payment

Card Number: 4111111111111111

Expiry: 12/30

CVV:123

Cardholder: must have more 2 character

Card dont have tail 0000

Back

Select Payment Method

Credit Card Number format is invalid. Please check your card details and try again.

TOTAL AMOUNT TO PAY

3,442,500 đ

Cash on Delivery

Pay when you take the car

Bank Transfer

Direct via Banking app

MoMo Wallet

Scan QR Code to pay

Credit / Debit Card

Visa, Master, JCB

4.2.2.8 Scrum Board

This is the SCRUM Board for Sprint 2

Q Search

+ Create

Spaces

PrivateHire Web-app

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Developn

Q Search board

1L

Filter

Cc

PHW-15 US10: Detailed Service Information and Availability (4 subtasks) SEARCHING & BOOKING SERVI... TO DO

PHW-18 US13: Online Trip Booking with 15% Discoun (5 subtasks) SEARCHING & BOOKING SERVI... TO DO

PHW-40 US16: Online Trip Booking Details: location, date and time (4 subtasks) SEARCHING & BOOKING SERVI... TO DO

PHW-41 US17: Vehicle Capacity and Size Selection (4 subtasks) SEARCHING & BOOKING SERVI... TO DO

PHW-21 US21: Booking Cancellation (24-Hour Rule) (6 subtasks) BOOKING MANAGEMENT & NOT... TO DO

PHW-26 US23: Secure Credit Card Payment (5 subtasks) PAYMENT TO DO

PHW-28 US25: Payment Verification via VISA Check (4 subtasks) PAYMENT TO DO

Everything else (no work items)

Figure 4.2.2.8.1: Sprint 2 SCRUM Board before the development process when showing only user stories

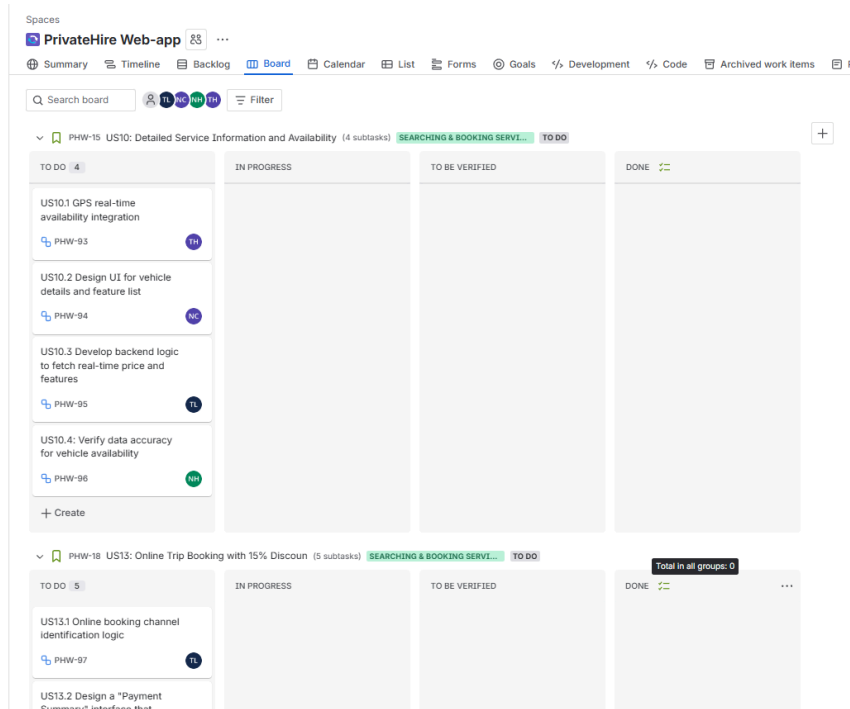


Figure 4.2.2.8.2: Sprint 2 SCRUM Board before the development process when showing tasks

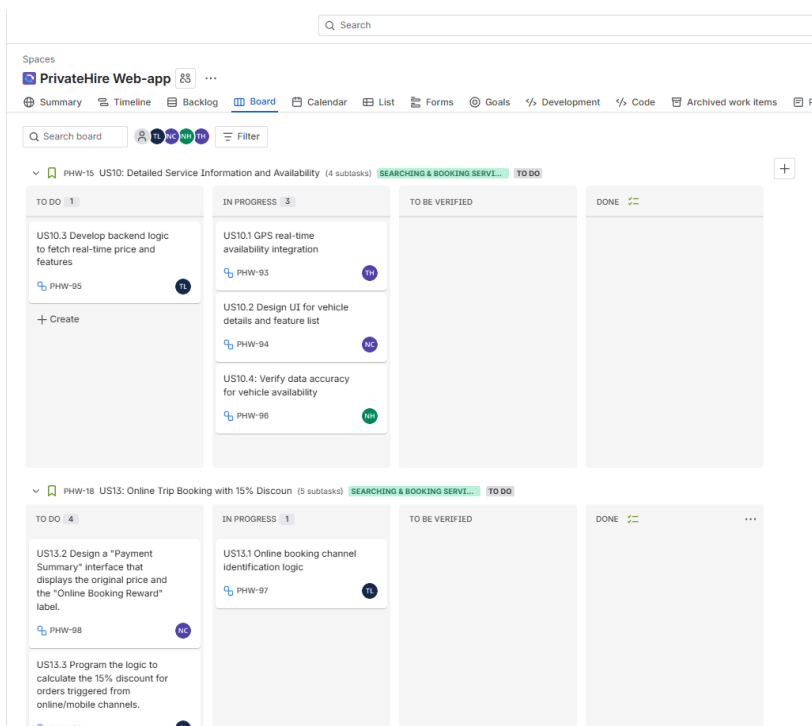


Figure 4.2.2.8.3: Sprint 2 SCRUM Board during the development process

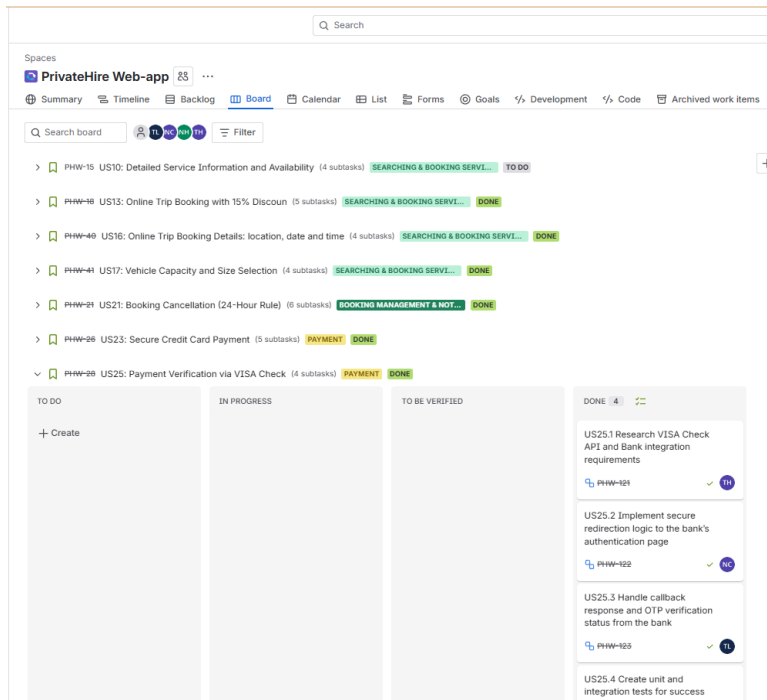


Figure 4.2.2.8.4: Sprint 2 SCRUM Board after the Sprint has concluded

4.2.2.9 Burndown charts

These are the burndown charts for Sprint 2

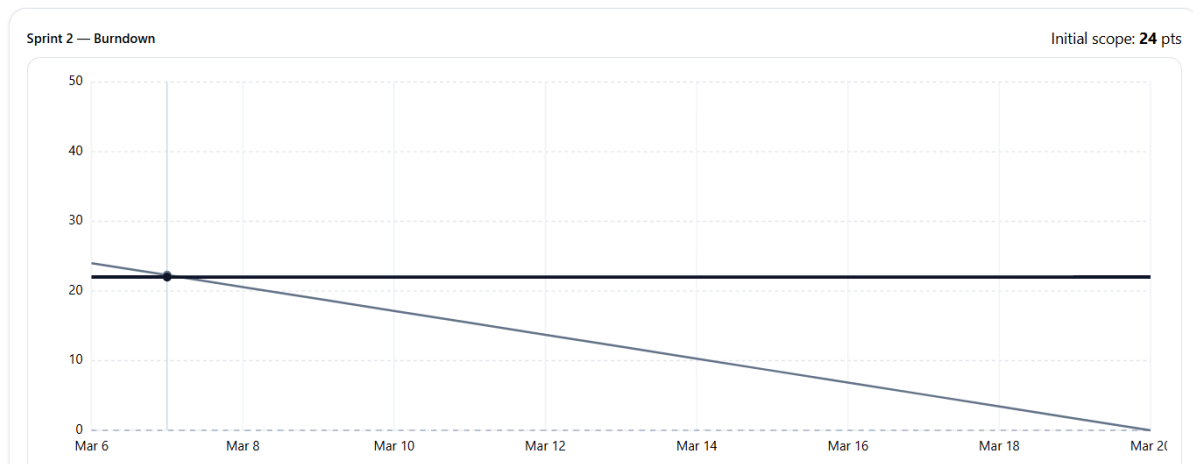


Figure 4.2.2.9.1: Sprint 2's Burndown chart up to March 7th

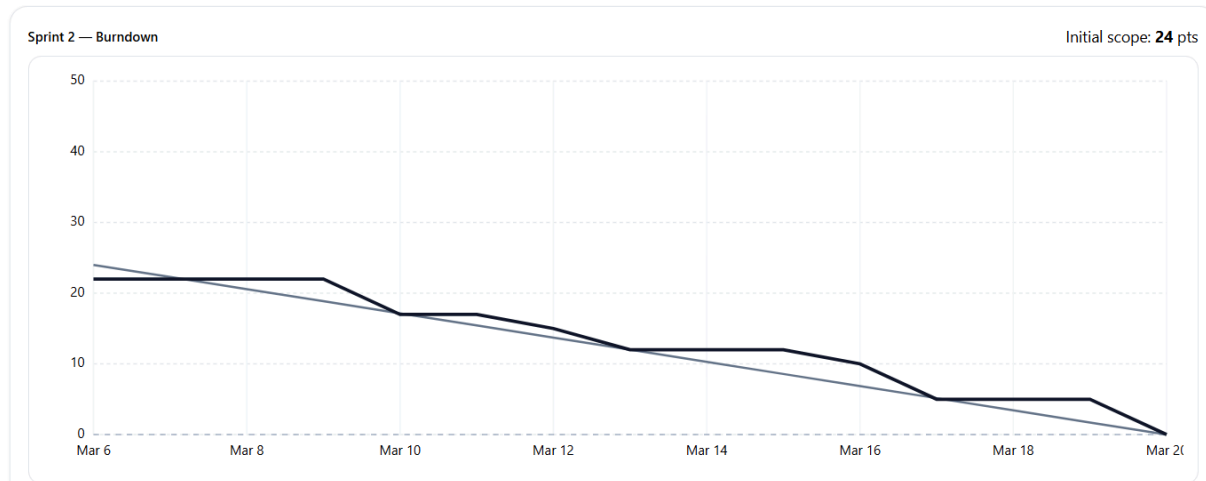


Figure 4.2.2.9.2: Sprint 2's Burndown chart at the end of the Sprint on March 20th

4.2.2.10 Meetings

❖ Daily meetings

Day	Focus Areas	Artifact status
1-2	UI/UX Design for Booking Forms (US16) and Payment UI (US23).	Scrum Board: Design tasks moved to Done.
4-5	Backend logic for price calculation and 15% discount (US13).	Burndown: Points dropped from 24 to 18.
7-8	VISA Check integration and OTP handling (US25).	Scrum Board: US10, 16, and 17 reached Done.
9-10	Regression testing and security validation.	All items moved to the 'Done' column.

❖ sprint review meetings

The team demonstrated the completed "Online Booking" lifecycle to stakeholders. Key usability enhancements included the following:

- Booking Validation: Preventing users from selecting past dates or identical pickup/drop-off points (US16).
- Payment Security: Providing clear UI feedback for successful or failed VISA cheque transactions (US25).
- Discount Visibility: Showing "Original Price" vs. "Discounted Price" on the final payment screen (US13).

❖ retrospective meetings

1. What Went Well

The adoption of standardised naming conventions from Sprint 1 significantly reduced refactoring time. Communication during daily scrums effectively resolved API connection issues early.

2. What Didn't Go Well

Estimating the complexity of the third-party bank redirection (US25) was tight, leaving little room for unexpected bugs.

3. Improvements for Sprint 3

Since Sprint 3 is a Hardening Sprint, the team will prioritize "Refactoring Mercilessly" and final documentation rather than adding major new features

❖ Summary Performance

The team delivered a potentially shippable increment that fulfils the core MVP requirements for private hire cars. We are well-prepared for the final demonstration.

5. Sprint Three

5.1 Sprint Planning and Sprint Backlog

5.1.1 Sprint Goal

The goal of Sprint 3 is to finalise the customer experience and system security for the PrivateHire Cars platform. This sprint focuses on implementing essential support and account management features, including forgotten password recovery (US03), user profile management (US05), and safety-critical text back notifications (US22). Additionally, the team will establish post-journey engagement through Receipt Generation (US27) and Star Ratings (US28). As the final phase, this sprint also serves as a hardening and release sprint, dedicated to system stabilisation, refactoring, and final documentation to ensure a "potentially shippable" product for the client.

5.1.2 Sprint "Definition of Done"

Sprint 3 is considered successfully completed when the following global criteria are met:

- ✓ User Story Fulfilment: All specific acceptance criteria for the selected User Stories (US03, US05, US22, US27, US28, US34, and US35) have been fully met and verified through testing.
- ✓ Technical Readiness: All new features are functional in the test environment, and unit/integration tests have passed with 100% success
- ✓ Regression Testing: The team has confirmed that the core functionalities from Sprints 1 and 2 (Registration, Booking, and VISA Check payment) remain fully operational after the integration of Sprint 3 code
- ✓ Scrum Board Accuracy: All user stories and their sub-tasks have been moved to the "Done" column on the Scrum/Kanban board
- ✓ Refactoring and Documentation: The codebase has been refactored mercilessly to ensure a simple, maintainable design, and all project manuals or release documentation are complete

✓ Product Owner Approval: The product owner (Mr Andrew Fowler) has reviewed the increment and officially accepted the deliverables as meeting the required business value for private hire cars.

5.1.3 Sprint Backlog

❖ Sprint Capacity Planning

Step 1: Key Variables	
	Team Size: 4 developer
	Sprint Length: 2 weeks (10 working days, excluding weekends)
	allocation: 3 Full-time developer (8h/day) and 1 Part-time developer
	Individual Availability: 100% available during scheduled hours
Step 2: Daily Capacity Calculation	
	Full-time developer 1: 8 hours/day
	Full-time developer 2: 8 hours/day
	Full-time developer 3: 8 hours/day
	Part-time developer 4: 4 hours/day
	Total Daily Team Capacity: $(3 \times 8h) + (1 \times 4h) = 28 \text{ hours/day}$
Step 3: Total Sprint Capacity	
	Formula: (Daily Team Capacity) $\times$ (Number of Working Days) Calculation: 28 hours/day $\times$ 10 days = 280 man-hours.
	including 40 hours allocated for Scrum activities such as sprint planning, daily Scrum, sprint review, and retrospective. Productive Time: 280-40 = 240 hours.
	Velocity per hour = Total Daily Team Capacity / 10 working days = 28 / 280 man-hours = 28 / 280 = 0.1
	Forecasted capacity: 240 $\times$ 0.1 = 24 story points

Derived from the productive time analysis, the team's forecasted capacity for this sprint is 24 story points. To match this capacity, the team carefully selected 5 user stories (US03, US05, US22, US27, US28, US34, and US35) for the sprint backlog, totalling exactly 17 points. This

consistency between the projected capacity and the actual workload ensures a realistic, non-overloaded sprint plan, thereby maximising the likelihood of delivering all committed work meeting the Definition of Done (DoD) by the sprint's conclusion.

User story ID	Title (user story)	Story Points	CR
US03	Forgotten Password Recovery	3	RQ1
US05	User Profile Management	2	RQ1
US22	text Back Notification for Driver Identification	5	RQ6
US27	Payment Confirmation and Receipt Generation	2	RQ8
US28	Journey Review and Star Rating	2	RQ9
US34	Live Chat Incident Support	2	RQ13
US35	Immediate Callback Support (Call Me Now)	1	RQ13
System Hardening		Release Prep	
Total: 17 points			

Having finalised the sprint backlog with a committed capacity of 17 story points, the team proceeded to break down these user stories into actionable technical tasks. The following section details the task breakdown required to fulfil the acceptance criteria and definition of done for each selected item.

5.1.4 Breaking down user stories included in the Sprint Backlog into Tasks

Title (User story)	task Description	task category	estimates (hours)
US03: Forgotten Password Recovery	US03.1 Design the password reset request link and the new password entry modal for the login interface.	Design	4
	US03.2 Develop backend logic to validate registered emails and generate secure, time-limited reset tokens.	code	5

	US03.3 Integrate the automated email system to deliver reset links to the user's registered address.	code	6
	US03.4 Create the secure "New Password" page with input validation for security standards.	code	4
	US03.5 Perform end-to-end testing of the password recovery flow from email request to login.	test	4
US05: Profile Management	US05.1 Design UI for user profile editing fields	Design	4
	US05.2 Implement backend update logic for user data	code	7
	US05.3 Functional test for profile data persistence	test	4
US22: text Back Notification for Driver Identification	US22.1 Design an email template to receive notifications.	Design	4
	US22.2 Send trusted emails to process automated text messages from a database and send them to customers.	code	8
	US22.3 Implement trigger logic to send accurate vehicle details to the user's email within 10-15 minutes of booking.	code	6
	US22.4 Develop SQL queries to retrieve detailed driver/vehicle information (Booking ID, Vehicle Name, Manufacturer, Number of Seats, Service Type, Booking Location, Arrival Location, Booking/Closing Time) in real time for email messages.	code	4
	US22.5 Conduct integration testing to verify SMS delivery and accuracy of car information.	test	6

US27: Payment Confirmation and Receipt Generation	US27.1 Create a printable HTML/PDF receipt template including journey details, a 15% discount, and the final cost. Receipt Template	Design	4
	US27.2 Develop logic to trigger receipt generation immediately after a successful bank verification (Visa Check).	code	7
	US27.3 Add "Print Receipt" and "Download PDF" buttons to the post-payment confirmation screen.	code	7
	US27.4 Perform end-to-end integration testing to ensure the receipt is generated correctly and only after a successful transaction	test	4
US28: Journey Review and Star Rating	US28.1 Design the feedback interface for users to select star ratings (1-5) and enter text comments	Design	5
	US28.2 Develop the database schema to store ratings for journeys, vehicles, and drivers	code	5
	US28.3 Implement the trigger to display the review prompt only after a journey status is "completed".	code	6
	US28.4 Testing: Verify stars are saved correctly and linked to the specific driver/car	test	2
US34: Live Chat Incident Support	US34.1 Integrate a real-time chat widget for customers to report incidents or ask queries	Design	4
	US34.2 Develop the backend messaging system to connect customer chats to the Call Centre staff dashboard	code	8
	US34.3 Create the Sales/Staff interface for monitoring and replying to active support sessions	code	6

	US34.4 Chat Integration Test	test	4
US35: Immediate Callback Support	US35.1 Add the "Call Me Now" button to the support/incident reporting page	Design	3
	US35.2 Implement an automated alert system that notifies Call Centre staff immediately of a callback request	code	5
	US35.3 Testing: Ensure the staff dashboard receives an audible or visual alert when a request is made.	test	2
Total: 138 hours			

❖ Justification for Task Estimation and Agile Thinking

For Sprint 3, the team committed to 17 story points, purposefully reducing the load from our 24-point velocity to focus on system hardening and release preparation. Following Scrum.org principles, we prioritise delivering a high-quality, stable product over maximum resource utilisation.

- Out of 240 available hours, we allocated 111 hours for core technical tasks (US03, US05, US22, US27, US28, US34, US35). The remaining 102 hours are dedicated to "slack time" for final quality assurance and XP engineering disciplines:
- Refactoring & Technical Debt Cleanup (25 hrs): "Refactoring Mercilessly" to optimise code structure and ensure the system is maintainable before release.
- Pair Programming on Support Modules (15 hrs): Collaborative coding on complex real-time features like Live Chat (US34) and automated SMS triggers.
- Strategic Buffer (6 hrs): Reserved for final impediment resolution during the daily scrums to maintain a sustainable pace
- Release Prep & Final Documentation (56 hrs): Adding "finishing touches" to the UI and creating final technical manuals for the handover.

Test Tracking List (40 hours)

Test Case ID	Test Type	Acceptance Criteria (AC)	Expected Result	Actual Result	Status	Notes	Test Date
TC35	UI/UX	US03.1 Design the password reset request	The password reset link and entry	As Expected	Pass		24/03

		link and the new password entry modal for the login interface.	Modals are correctly designed and visible on the interface.				
TC36	Security	US03.2 Develop backend logic to validate registered emails and generate secure, time-limited reset tokens.	The system generates secure, time limited tokens only for valid, registered email addresses.	As Expected	Pass		24/03
TC37	Functional	US03.3 Integrate the automated email system to deliver reset links to the user's registered address.	The automated system successfully delivers the reset link email to the correct user.	Not As Expected	Fail	The password reset confirmation email was not sent to the correct user. Fixed.	25/03
TC38	Unit	US03.4 Create the secure "New Password" page with input validation for security standards.	The "New Password" page is created with functional validation for complexity and security.	As Expected	Pass		25/03
TC39	UI/UX	US05.1 Design UI for user profile editing fields	The user profile editing fields are designed with an intuitive and clean layout.	As Expected	Pass		26/03

TC40	Database	US05.2 Implement backend update logic for user data	Backend logic correctly processes and saves updated user information to the database.	Not As Expected	Fail	The database did not receive the updated user information because a column was missing during the database construction process. Fixed.	26/03
TC41	UI/UX	US22.1 Design an email template to receive notifications.	Email users have received a warning notification.	As Expected	Pass		27/03
TC42	Database	US22.2 Send trusted emails to process automated text messages from a database and send them to customers.	Retrieve data from pre-stored messages in the database and successfully process and send automated messages.	As Expected	Pass		27/03
TC43	Logic	US22.3 Implement trigger logic to send accurate vehicle details to the user's email within 10-15 minutes of booking.	The system activates and sends accurate vehicle details to the user's email 10-15 minutes after booking.	As Expected	Pass		27/03
TC44	Database	US22.4 Develop SQL queries	The SQL query retrieves	As Expected	Pass		29/03

		to retrieve detailed driver/vehicle information (Booking ID, Vehicle Name, Manufacturer, Number of Seats, Service Type, Booking Location, Arrival Location, Booking/Closing Time) in real time for email messages.	the exact vehicle details (Booking ID, Vehicle Name, Manufacturer, Number of Seats, Service Type, Booking Location, Arrival Location, Booking/Closing Time) for the email content.				
TC45	UI/UX	US27.1 Create a printable HTML/PDF receipt template including journey details, a 15% discount, and the final cost. Receipt Template	The receipt template correctly includes all journey details, discounts, and final pricing.	As Expected	Pass		29/03
TC46	Logic	US27.2 Develop logic to trigger receipt generation immediately after a successful bank verification (Visa Check).	Logic successfully triggers receipt generation only after a successful Visa Check.	As Expected	Pass		29/03
TC47	Functional	US27.3 Add "Print	"Print" and "Download"	Not As Expected	Fail	The "print" and	30/03

		Receipt" and "Download PDF" buttons to the post-payment confirmation screen.	buttons are functional on the confirmation screen.			"download" buttons work, but the PDF file contains no information. Fixed.	
TC48	UI/UX	US28.1 Design the feedback interface for users to select star ratings (1-5) and enter text comments	The feedback interface allows users to select star ratings and provide comments easily.	Not As Expected	Fail	Lack of rating symbol labels. Fixed.	31/03
TC49	Database	US28.2 Develop the database schema to store ratings for journeys, vehicles, and drivers	The database schema is created to store ratings for journeys, vehicles, and drivers.		Pass		31/03
TC50	Unit	US28.3 Implement the trigger to display the review prompt only after a journey status is "completed".	The review prompt is triggered and displayed only for "completed" journeys.	As Expected	Pass		03/04
TC51	UI/UX	US34.1 Integrate a real-time chat widget for customers to report incidents or ask queries	The real-time chat widget is integrated and accessible for customer inquiries.	As Expected	Pass		03/04

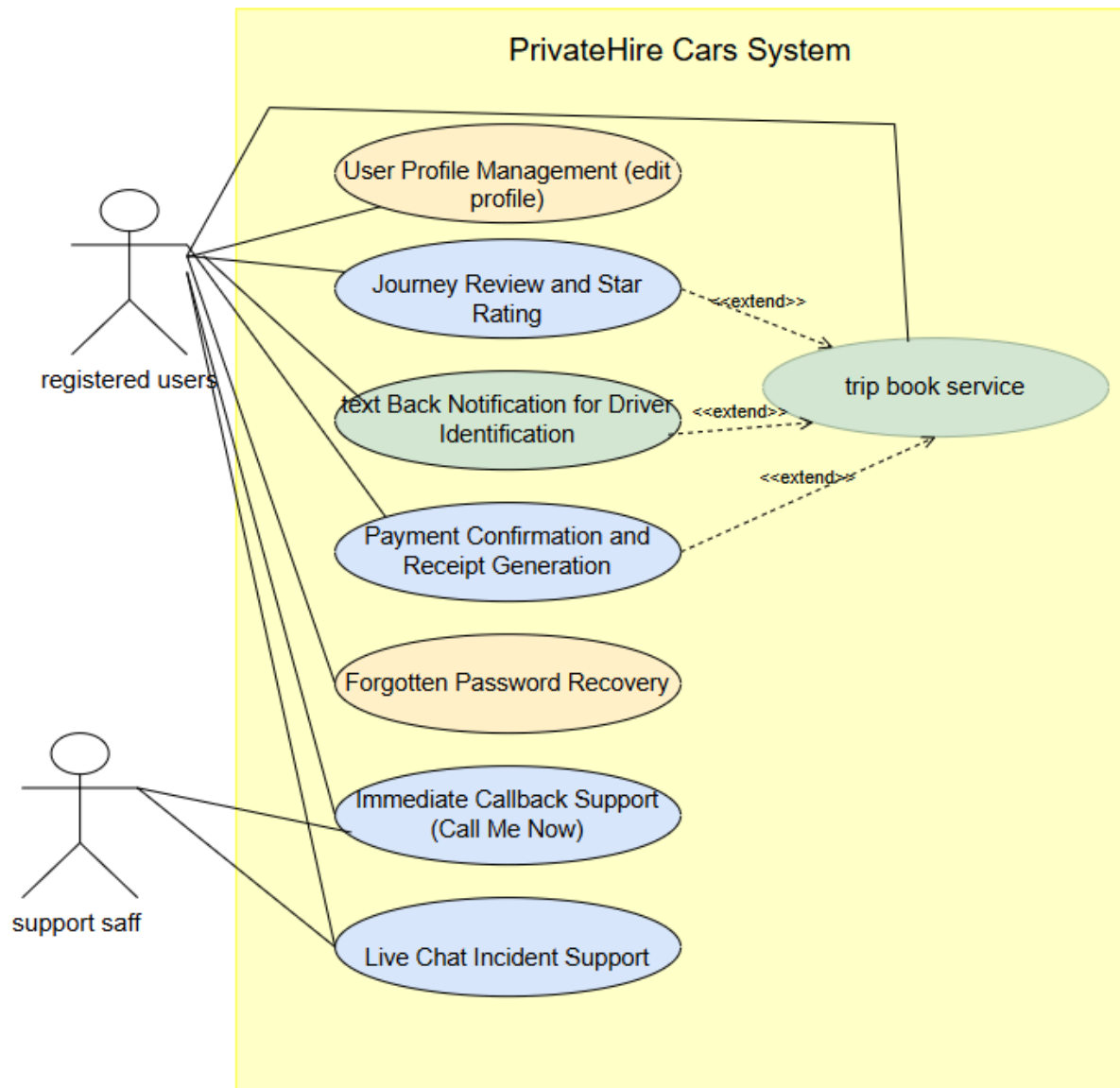
TC52	Unit	US34.2 Develop the backend messaging system to connect customer chats to the Call Centre staff dashboard	The backend correctly connects customer chat messages to the staff dashboard.	As Expected	Pass		05/04
TC53	Functional	US34.3 Create the Sales/Staff interface for monitoring and replying to active support sessions	The staff interface is functional for monitoring and replying to live chat sessions.	As Expected	Pass		05/04
TC54	UI/UX	US35.1 Add the "Call Me Now" button to the support/incident reporting page	The "Call Me Now" button is correctly placed and visible on the support page.		Pass		06/04
TC55	Functional	US35.2 Implement an automated alert system that notifies Call Centre staff immediately of a callback request	The automated system notifies staff immediately when a callback request is made.	Not As Expected	Fail	Customer call data was not being sent to the call center staff due to an incorrect MySQL table path. Fixed.	07/04

5.2 Sprint Cycle

5.2.1. Modelling

During Sprint 3, the team finalised the use-case diagram by integrating advanced functionalities into the core framework established in Sprints 1 and 2. Key enhancements included Forgotten Password Recovery (US03) and User Profile Management (US05) to strengthen account security and user control. Furthermore, the team modelled the Text Back service (RQ6) for secure driver identification, the Journey Review and Star Rating system (RQ9), and real-time support channels, including live chat (US34) and immediate callback (US35). This final modelling phase ensures the system meets the Definition of Done and is fully prepared for the system hardening and release phase

5.2.1.1. Use-Case Diagram



5.2.2 Development

5.2.2.1 US03: Forgotten Password Recovery

User Story: As a user, I want to retrieve or reset my forgotten password, so that I can regain access to my account and continue using the booking services

Forgot Password Step 1: Enter your email to receive a 6-digit OTP code. Email * tan0979876976@gmail.com Send OTP Already have an OTP? Go to OTP confirmation Back to Login	Car Booking System <tan0979876976@gmail.com> đến tôi ▾ Hello Minh Hoang, Your password reset OTP is: 042670 This OTP is valid for 5 minutes.	Forgot Password OTP has been sent successfully. Please check your inbox. Step 2: Enter your OTP code to confirm your identity. Email * tan0979876976@gmail.com OTP Code * 042670 Confirm OTP Need a new OTP? Back to send OTP Back to Login
Forgot Password OTP verified successfully. Please set your new password. Step 3: Set your new password. Email * tan0979876976@gmail.com New Password * ***** Confirm New Password * Tfcvg@12345. Set New Password Need to re-confirm OTP? Back to OTP confirmation Back to Login	Login Password reset successful Email * tan0979876976@gmail.com Password * ***** ☐ Remember me Forgot password? Login Don't have an account? Register now	1. enter email 2. enter otp 3. change new password

5.2.2.2 US05: User Profile Management

User Story: As a user, I want to edit my profile information so that I can keep my personal information up-to-date.

Home Browse Cars My Bookings **My Profile** Need Help?

My Profile

Email
tan0979876976@gmail.com

Full Name *
Minh Hoang

Phone Number *
0907495718

Address
47/SH xã Học Môn

Member since 19/03/2026 12:20

SAVE PROFILE

5.2.2.3 US22: Text Back Notification for Driver Identification

User Story: As a user, I want to receive a text message 10-15 minutes before arrival with the car's registration number, colour, and make so that I can safely identify my driver and avoid bogus vehicles.

Booking Completed

Thank you for choosing Born Car. Your booking is successfully created.

Booking ID: #39

CUSTOMER

Minh Hoang

EMAIL

tan0979876976@gmail.com

CAR

Honda City RS

SERVICE TYPE

With Driver

PICK-UP TIME

16 Apr 2026 - 10:00

DROP-OFF TIME

18 Apr 2026 - 09:00

PICK-UP AREA

Tan Binh District

PICK-UP LANDMARK

aaaaaaa

DROP-OFF AREA

Binh Thanh District

DROP-OFF LANDMARK

aaaaaaa

PAYMENT METHOD

Cash

STATUS

confirmed

Total Payment: **2,700,000 VND**

5.2.2.4 US27: Payment Confirmation and Receipt Generation

User Story: As a user, I want to receive a confirmation email and a receipt after payment so that I have proof of the transaction



Born Car <tan0979876976@gmail.com>
đến tôi ▾

Booking Confirmed

Hello **Minh Hoang**,

Your booking has been placed successfully. Here are your booking details:

- **Booking ID:** #38
- **Car:** Honda City RS
- **Pick-up:** 12/04/2026 09:00
- **Drop-off:** 15/04/2026 09:00
- **Service Type:** With Driver
- **Pick-up Area:** District 3
- **Pick-up Landmark:** aaaaaaaa
- **Drop-off Area:** Tan Binh District
- **Drop-off Landmark:** aaa
- **Payment Method:** Cash
- **Total Price:** 4,050,000 VND

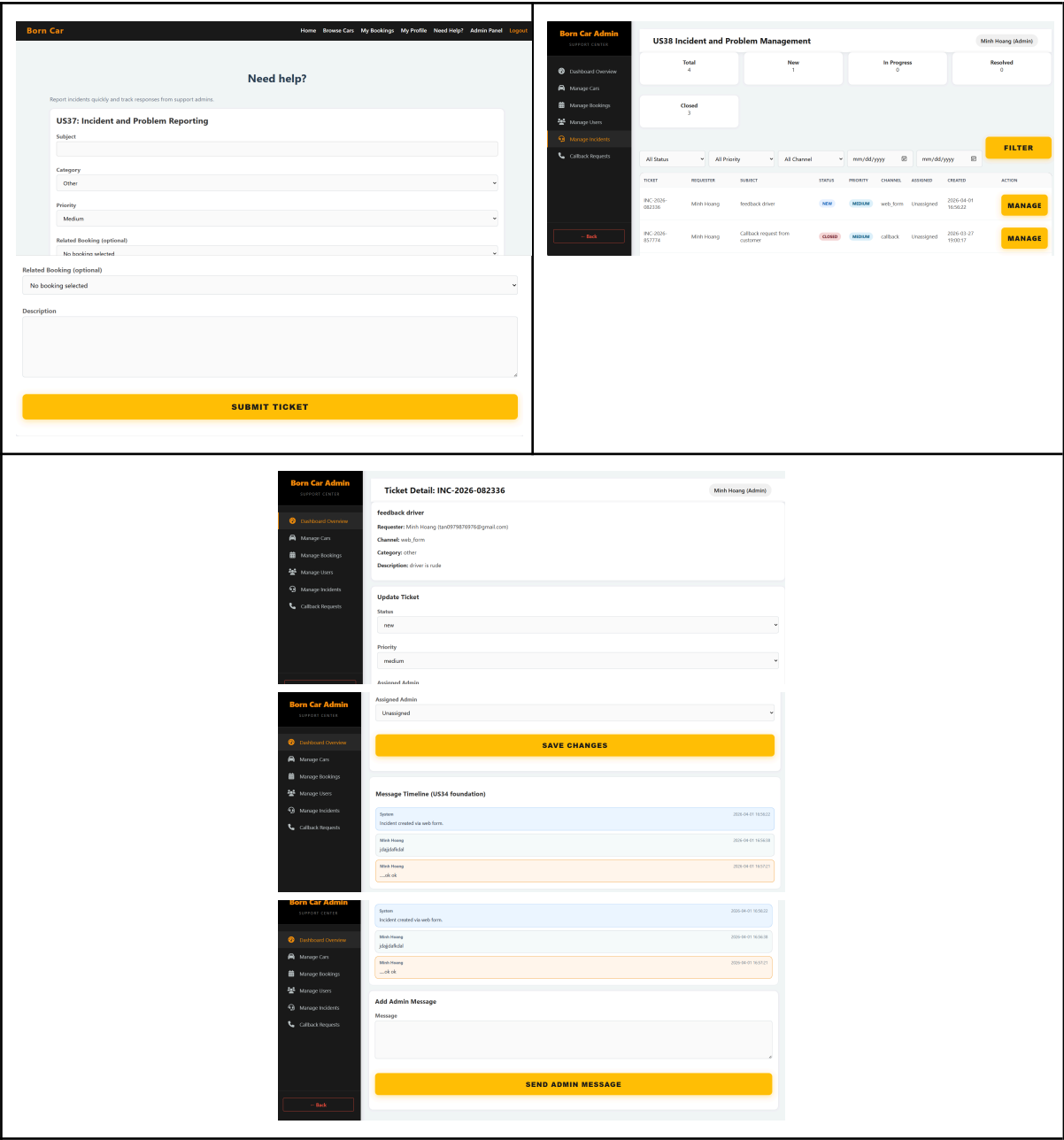
Thank you for choosing Born Car.

5.2.2.5 US28: Journey Review and Star Rating

User Story: As a user, I want to provide feedback and a rating so that I can share my experience and help improve services.

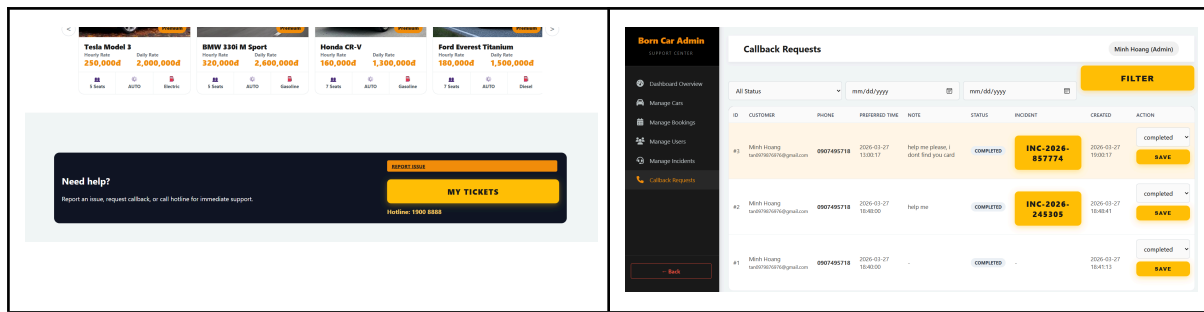
5.2.2.6 US34: Live Chat Incident Support

User Story: As a user, I want to ask for support on an incident via chat so that I can resolve issues quickly



5.2.2.7US35: Immediate Callback Support (Call Me Now)

User Story: As a user, I want a "Call Me Now" feature so that I can request an immediate callback from support for urgent issues



5.2.2.8 Scrum Board

+ Create
Add payment detail

Spaces

PrivateHire Web-app

Summary
Timeline
Backlog
Board
Calendar
List
Forms
Goals
Development
Code

TL
NC
NH
TH

Filter

Complete sprint

PHW-9
US03: Forgotten Password Recovery (5 subtasks)
ACCOUNT MANAGEMENT & AUT...
TO DO

PHW-37
US05: User Profile Management (3 subtasks)
ACCOUNT MANAGEMENT & AUT...
TO DO

PHW-22
US22: Text Back Notification for Driver Identification (5 subtasks)
BOOKING MANAGEMENT & NOT...
TO DO

PHW-30
US27: Payment Confirmation and Receipt Generation (5 subtasks)
PAYMENT
TO DO

PHW-31
US28: Journey Review and Star Rating (4 subtasks)
CUSTOMER ENGAGEMENT & LO...
TO DO

PHW-48
US34: Live Chat Incident Support (4 subtasks)
SUPPORT
TO DO

PHW-49
US35: Immediate Callback Support (Call Me Now) (3 subtasks)
SUPPORT
TO DO

Everything else (no work items)

5

Q Search

Spaces

PrivateHire Web-app

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

Archived work item

Q Search board

TL NC NH TH

Filter

PHW-22 US22: Text Back Notification for Driver Identification (5 subtasks) **BOOKING MANAGEMENT & NOT...** **TO DO**

PHW-30 US27: Payment Confirmation and Receipt Generation (5 subtasks) **PAYMENT** **TO DO**

TO DO 5

US27.1 Create a printable HTML/PDF receipt template including journey details, a 15% discount, and the final cost.
PHW-138 NC

Receipt Template
PHW-139 TL

US27.2 Develop logic to trigger receipt generation immediately after a successful bank verification (Visa Check).
PHW-140 TL

US27.3 Add "Print Receipt" and "Download PDF" buttons to the post-payment confirmation screen.
PHW-141 TH

US27.4 Perform end-to-end integration testing to ensure the receipt is generated correctly and only after a successful transaction
PHW-142 NH

IN PROGRESS

TO BE VERIFIED

DONE 2

Q

Search

+ Create

Spaces

PrivateHire Web-app

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

Archived work items

Pages

Shortcuts

Q

Search board

TL

NC

NH

TH

Filter

PHW-37

US05: User Profile Management

(3 subtasks)

ACCOUNT MANAGEMENT & AUT...

TO DO

TO DO

IN PROGRESS

TO BE VERIFIED

DONE 3

US05.3 Functional test for profile data persistence

PHW-132

✓

NH

PHW-22

US22: Text Back Notification for Driver Identification

(5 subtasks)

BOOKING MANAGEMENT & NOT...

TO DO

TO DO 1

US22.4 Develop SQL queries to fetch real-time driver/vehicle details (registration, colour, and make) for the SMS body.

PHW-136

TL

IN PROGRESS 3

US22.2 Send trusted emails to process automated text messages from a database and send them to customers.

PHW-134

TH

US22.3 Implement trigger logic to send accurate vehicle details to the user's email within 10-15 minutes of booking.

PHW-135

TL

US22.5 Conduct integration testing to verify SMS delivery and accuracy of car information.

PHW-137

NH

TO BE VERIFIED 1

US22.1 Design an email template to receive notifications.

PHW-133

NC

DONE

...

84

Q Search

Spaces

PrivateHire Web-app

...

Summary

Timeline

Backlog

Board

Calendar

List

Forms

Goals

Development

Code

Archived work items

Q Search board

TL

NC

NH

TH

Filter

PHW-22

US22: Text Back Notification for Driver Identification

(5 subtasks)

BOOKING MANAGEMENT & NO...

DONE

PHW-36

US27: Payment Confirmation and Receipt Generation

(5 subtasks)

PAYMENT

DONE

PHW-31

US28: Journey Review and Star Rating

(4 subtasks)

CUSTOMER ENGAGEMENT & LO...

DONE

PHW-46

US34: Live Chat Incident Support

(4 subtasks)

SUPPORT

DONE

PHW-49

US35: Immediate Callback Support (Call Me Now)

(3 subtasks)

SUPPORT

DONE

TO DO

+ Create

IN PROGRESS

TO BE VERIFIED

DONE 3

US35.1 Add the "Call Me Now" button to the support/incident reporting page

PHW-151

✓

TL

US35.2 Implement an automated alert system that notifies Call Centre staff immediately of a callback request

PHW-152

✓

TH

US35.3 Testing: Ensure the staff dashboard receives an audible or visual alert when a request is made.

PHW-153

✓

NH

Everything else (no work items)

5.2.2.9 Burndown charts

Sprint 3 — Burndown

Initial scope: 17 pts

Date	Actual Progress (pts)	Remaining (pts)
Mar 24	17	0
Mar 25	14	3
Mar 26	14	3
Mar 27	12	5
Mar 28	12	5
Mar 29	12	5
Mar 30	10	7
Mar 31	9	8
Apr 1	9	8
Apr 2	4	13
Apr 3	4	13
Apr 4	4	13
Apr 5	4	13
Apr 6	4	13
Apr 7	4	13

Figure 5.2.2.9.1: Sprint 3's Burndown chart up to April 2nd

85



Figure 5.2.2.9.2: Sprint 3's Burndown chart at the end of the Sprint on April 7th

5.2.2.10 Meetings

❖ Daily meetings

Day	Focus Areas	Artifact status
1-2	Implementing Password Recovery (US03) and Profile Editing (US05).	Scrum Board: Account management tasks moved to Done.
4-5	Developing the Text Back service (US22) and Support features (US34, US35).	Burndown: Dropped from 17 to 10 points.
7-8	System Hardening: "Refactoring Mercilessly" to improve code structure.	Scrum Board: US27 and US28 reach 100% completion.
9-10	Final Release Prep: Completing user manuals and testing the full business life cycle.	All items moved to the 'Done' column. Ready for Demo.

❖ sprint review meetings

This served as our final dry run before the tutor demonstration. We "sold" the system by showing the following:

- Safety Logic (US22): Demonstrating how the text-back service eliminates the chance of being picked up by a bogus driver.
- Customer Feedback (US28): Showing how users can rate their journey and driver to improve future satisfaction.
- Support Integration (US34/35): Presenting the "Call Me Now" and Live Chat features for urgent incident handling.

❖ retrospective meetings

What Went Well: The lighter point load (17 SP) allowed the team to achieve a high level of code quality through pair programming on the support modules.

1. What Didn't Go Well

Coordinating the real-time notification logic (US22) with the simulated GPS system was technically challenging and took longer than expected.

2. Final Evaluation

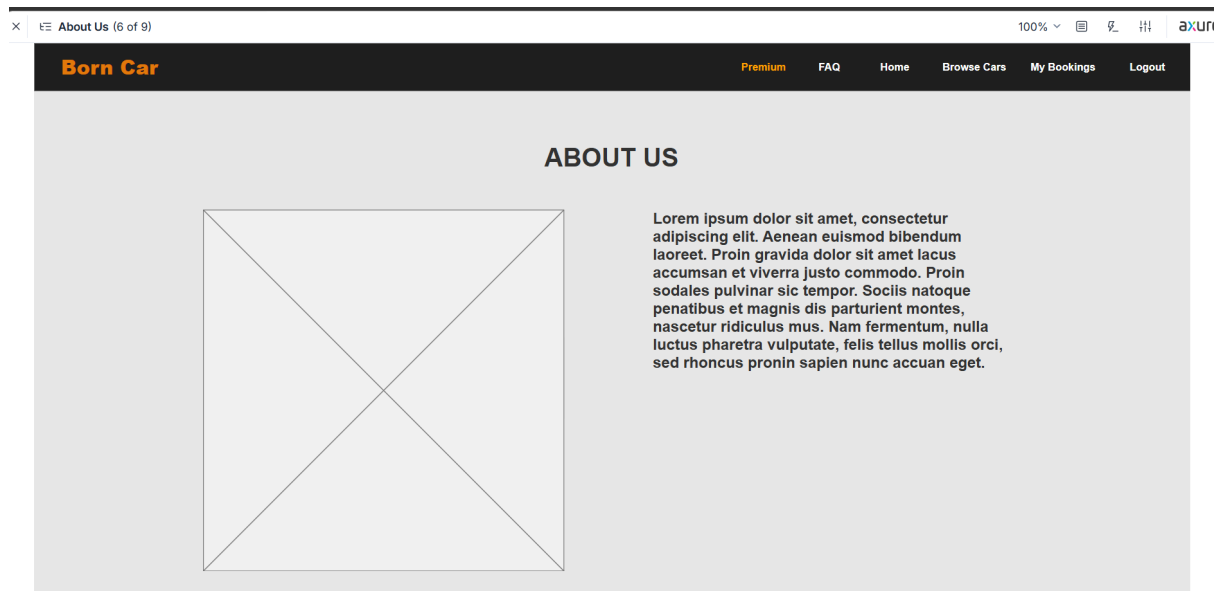
The team successfully delivered a potentially shippable increment. We have evolved from individual developers into a self-organising Scrum team capable of maintaining a sustainable pace while delivering business value for private hire cars.

3. Final Status

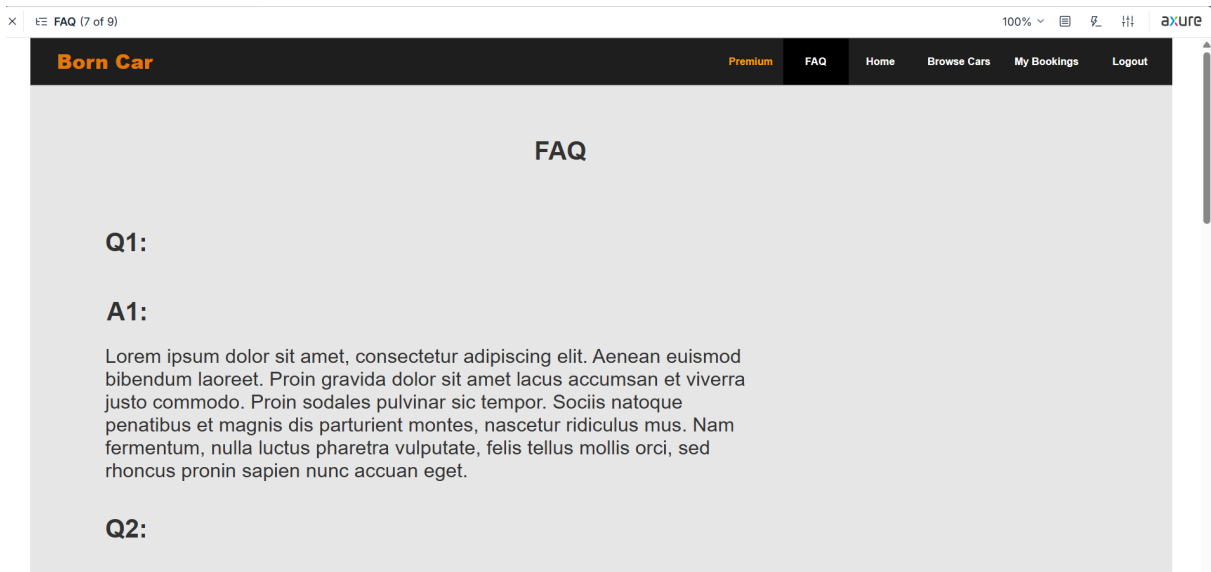
The prototype system now supports the full business life cycle, from registration and secure booking to post-trip reviews and customer support, meeting the Minimum Viable Product standards.

Below are some unfinished prototypes in the backlog

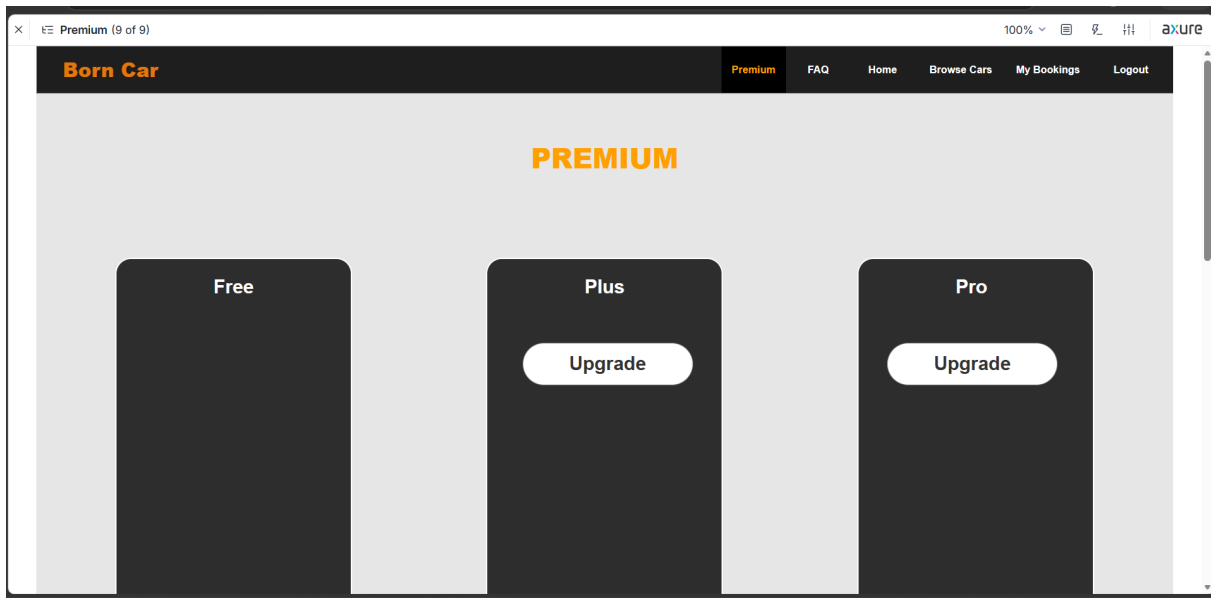
US19 About Us

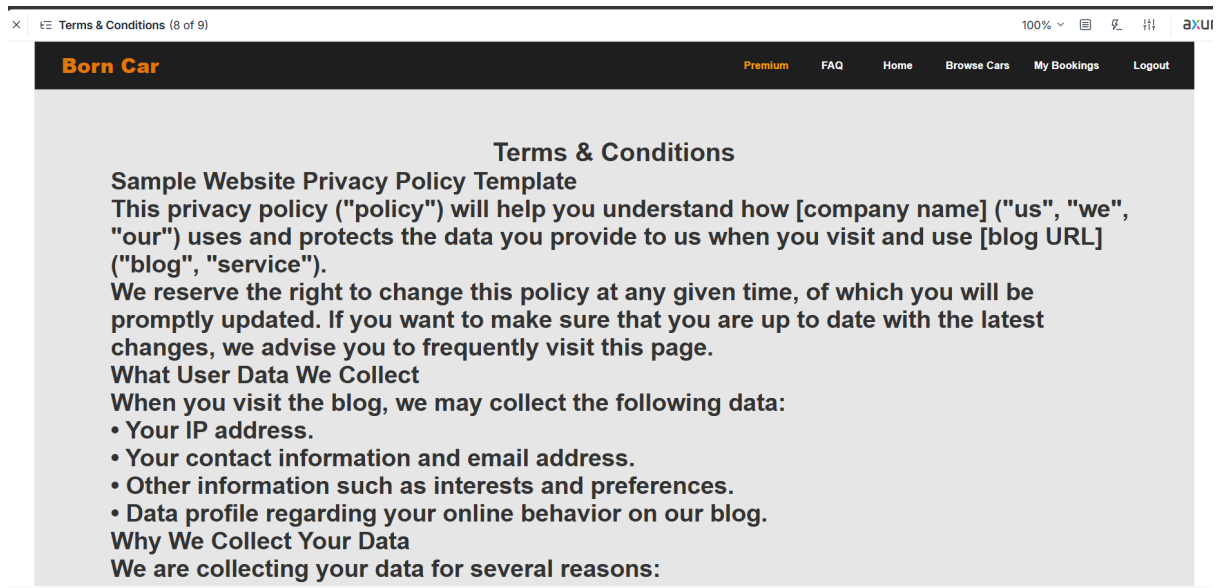


us18 FAQ



US29: Membership and Loyalty Upgrades





6. Individual Report – Reflection and Evaluation

6.1. Discussion of the course and what was learnt

This course provided a deep dive into Agile/Scrum, focusing on iterative delivery and maintaining a sustainable pace. I mastered managing the Product Backlog as a "living artifact," converting requirements into User Stories and Acceptance Criteria. Additionally, XP rituals (Pair Programming) and Planning Poker were essential for technical discipline, helping us decompose EPICs into manageable, high-quality tasks.

In addition, Scrum ceremonies like Daily Scrums and Retrospectives supported continuous feedback, while User Stories and Acceptance Criteria ensured value-driven prioritization. This lifecycle reinforced collaboration and transparency, proving that incremental delivery is essential for project success.

In this project, I spent most of my time focusing on managing team capacity during Sprint Planning, finalizing technical documentation and burndown charts, and refining the Product Backlog as a living artifact, facilitating Daily Scrums to remove team impediments, and developing the prototype's core user interface through Pair Programming to ensure every increment met our strict Definition of Done

6.2. Discussion of teamwork and assessment of team member

While project roles were clearly distributed, the primary obstacle was inconsistent communication, which occasionally led to task misalignment. Early stages were marked by limited engagement, hindering collective awareness and project synchronization.

To mitigate this, I facilitated more rigorous team interactions in the later phases, mandating frequent updates and active participation in all discussions. This pivot significantly enhanced our collaborative synergy and alignment. Although technical contributions were sufficient to ensure a successful delivery, this experience underscores that proactive communication is just as critical as technical skill in maintaining team velocity.

6.3. Sprint Retrospective

Our Sprints consistently delivered incremental value, with regular ceremonies providing the necessary structure for transparency. However, communication gaps initially compromised the accuracy of our Sprint Planning and task estimation.

We observed a direct correlation between increased engagement and improved requirement clarity. The Retrospective served as a vital correction tool, allowing us to transition from fragmented work to a more coordinated flow. For future iterations, I recommend a stricter adherence to communication protocols and refined estimation techniques to optimize workflow efficiency and ensure sustained team performance.

7. Reference

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Declaration of AI Use

Please append this page to the end of your assignment when you have used AI during the process of undertaking the assignment to acknowledge the ways in which you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – NO
- To explain concepts – NO
- To support my use of language – YES
- To summarise the following articles/resources: